

Theories of Bank Heterogeneity, Liquidity, Credit Risk and Monetary Shocks: A Literature Review^{*}

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Abstract

This paper reviews the theoretical and quantitative macroeconomic literature on bank heterogeneity. In the first part I present facts on the distribution of banks and I motivate own theoretical study of heterogeneous banks and monetary policy uncertainty by documenting the evidence of unconventional monetary policies in heterogeneously affecting banks according to their characteristics. I then discuss the specific features of the current state of heterogeneous banks theories, the ways they differ and what I believe are their limitations. I complement the analysis by empirically analysing the consequences of monetary shocks on the rebalancing mechanism of banks' assets between liquid and illiquid assets. I find a 10.9 and 10.1 percent reduction in illiquid assets growth for banks with more liquid assets pre and post the financial crisis, respectively. The results suggest an important role for deposits that lead to a reduction in illiquid assets growth that is 8.8 and 8.6 percentage points smaller, pre and post-financial crisis. The purpose of this paper is to raise consciousness of the importance of embedding heterogeneity in the financial sector of macroeconomic theories that aim to provide positive and normative analysis of monetary policies measures and to propose suggestions for further research.

Keywords: Banks' regulations, financial stability, market power, liquidity management, spatial banks

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1 Introduction

During the last three decades a new class of models in macroeconomics have been developed in order to provide a better understanding of the economy. This class of models are heterogeneous agents models with incomplete markets, that address serious disparities in terms of income and wealth, matters of inequality and distributional consequences. The literature has been focussing mainly on the heterogeneity to the households side and to a some extent to the firm side. The first generation of heterogeneous agents model dates back to more than three decades starting with the workhorse models of [Bewley \(1986\)](#), [Imrohoroglu \(1989\)](#), [Huggett \(1993\)](#) and [Aiyagari \(1994\)](#), followed by [Ríos-Rull \(1995\)](#), [Aiyagari and Gertler \(1999\)](#) and [Carroll et al. \(2021\)](#). These works use incomplete market models to develop the notion of an impossibility to insure against idiosyncratic risk in the consumer's decision problem. Since then a growing literature of heterogeneous agents would have culminated into the second generation of heterogeneous agents models. This generation of models distinguish themselves by having combined aggregate shocks, such as monetary and fiscal policy shocks, with incomplete markets subject to idiosyncratic risk.¹

The turmoils in the financial market after the global financial crisis and the resulting extensive empirical evidence, made clear the need for macroeconomists to start thinking reviewing the role of the financial sector and, in particular the banking sector, into macroeconomic models. The unconventional monetary policies undertaken by central banks were aimed to provide access to funds and liquidity to banks at a lower cost. They were designed to counteract the fall in aggregate demand and to encourage a credit expansion from banks in order to support an increase in households borrowing and spending and firms investments. However, these unconventional policies caused a response from banks that was non-identical in different dimensions. While the majority of countries relied on asset market purchases programmes, only a few monetary authorities such as the European Central Bank, Denmark Nationalbank, Sveriges Riskbank, the Bank of Japan and Swiss National Bank allowed for the joint combination of negative interest rates and asset market purchases programmes.²

The scope of this paper is fourfold. First, it documents facts on the distribution of banks in Section 2. Second, it provides a review of the literature on three strands of studies: the empirical literature on heterogeneous banks, the literature on monetary shocks with interest rates in low or negative region in Sections 3 and 4, and the literature on heterogeneous banks in macroeco-

¹Notably examples in a New Keynesian context include [Oh and Reis \(2012\)](#), [Kaplan and Violante \(2014\)](#), [Buera and Moll \(2015\)](#), [Werning \(2015\)](#), [Gornemann et al. \(2016\)](#), [Krueger et al. \(2016\)](#), [McKay et al. \(2016\)](#), [Achdou et al. \(2022\)](#), [Challe et al. \(2017\)](#), [den Haan et al. \(2018\)](#), [Guerrieri and Lorenzoni \(2017\)](#), [Kaplan et al. \(2018\)](#), [Nuño and Moll \(2018\)](#), [Auclert \(2019\)](#), [Bayer et al. \(2019\)](#), [Straub \(2019\)](#), [Alves et al. \(2020\)](#), [Bayer et al. \(2023\)](#), [Broer et al. \(2020\)](#), [Faccini et al. \(2026\)](#), [Nuño and Thomas \(2022\)](#), [Ravn and Sterk \(2020\)](#), [Luetticke \(2021\)](#), [McKay and Reis \(2021\)](#). A good guide on heterogeneous agents, although a little out of date, is present in [Güvenen \(2011\)](#).

²Asset Market Purchases Programmes using US banks level data were studied, amongst many other studies, by [Rodnyansky and Darmouni \(2017\)](#) who document cross-sectional heterogeneity amongst commercial banks finding that banks with more mortgage backed assets expanded more their lending after the quantitative easing programmes, and [Chakraborty et al. \(2020\)](#) find similar results.

nomic models in Section 5. Six general themes are considered: the empirical marginal propensity to lend of individual banks, the heterogeneous transmission of unconventional monetary policies, heterogeneous banks models in addressing liquidity risk, heterogeneous banks models in addressing credit risk, heterogeneous banks in bank runs models, heterogeneous banks models into complete markets economies.

Third, I provide empirical estimates on the rebalancing mechanism of assets following a monetary policy change. I use data from the Consolidated Report of Condition and Income from the Federal Reserve Bank of Chicago on US commercial banks, collected from [Wharton Research Data Services \(2023\)](#). Using different measures for banks' liquid assets, the results point to a restrictive effect on banks' illiquid assets which is statistically significant. The results are presented in Section 6. Fourth, I propose unanswered research questions for which I present a thesis of bank heterogeneity concerned with idiosyncratic and aggregate time variation in the stochastic volatility process. The former considers stochastic volatility in the idiosyncratic process of the individual heterogeneous bank problem whose time variation impacts the cross sectional distribution of banks directly, the latter affects the aggregate stochastic process such as monetary shocks and enters a bank decision problem indirectly. In both cases the volatility of uncertainty—the standard deviation of the stochastic volatility innovation—affects banks decision rules since the variance of the stochastic process becomes a relevant state variable. The specific source of volatility being considered matters for the heterogeneous effects of monetary policy transmission and the redistributational consequences to banks. These proposals are presented in Section 7.

In reviewing the theoretical literature of heterogeneous banks I focus extensively on two studies, [Bianchi and Bigio \(2022\)](#) and [Corbae and D'Erasmus \(2021\)](#) for two reasons. First, they incorporate bank heterogeneity through two different lenses. In [Bianchi and Bigio \(2022\)](#) heterogeneous banks are incorporated in an intrinsic way and heterogeneity disappears in the aggregate, whilst [Corbae and D'Erasmus \(2021\)](#) allows for an ex-post treatment of heterogeneous banks and the model has been taken as a reference for some other studies with heterogeneous banks that are presented in this literature.

Second, both papers provide theoretical frameworks that can be further explored. Therefore, I explain in detail the assumptions of these models, their consequences in their frameworks and the merits for their modelling choices. Then I address what I believe are their limitations, the restrictions that their assumptions pose and how these models can be further investigated.

The presence of empirical facts and evidence documenting a significant size distribution in the US banking system goes back to [Boyd and Gertler \(1994\)](#), [Ennis \(2001\)](#) and [Ennis and Malek \(2005\)](#). These studies have pointed towards the dominance of a few large banks owning more than 50% of the asset distribution. In recent years, the US banking market has become even more concentrated leading the regulator to denominate these few large banks as Systemically Important Financial Institutions (hereafter, SIFIs). However, what is missing from

the current theoretical literature is a positive theory that can rationalise the existence of these large banks, SIFIs, as an equilibrium outcome of the model and the distributional effects of monetary policies, in which by equilibrium outcome I mean the existence of a non degenerate distribution characterising the distribution of all banks observed in the banking sector. In addition, banks' behaviors in terms of willingness to lend and their motive to own liquidity is affected by the regulatory environment regarding the requirements they have to face in owning loans and liquidity. Therefore, understanding whether the behaviors of SIFIs are different from the behaviors of small banks in a time in which strong rules on their liquidity balances have been imposed after the financial crisis, has not been addressed by the theoretical literature. Even less clear is the discipline of the theory in explaining the empirically documented heterogeneous transmission of negative rates, as well as conventional monetary policies distributional effects on banks.

I present models that, despite consisting of heterogeneous banks, still deliver a representative bank or set of banks without a distribution to keep track, thereby representing a limit for the macroeconomic implications that can be drawn upon, especially in an environment in which changing the size of the idiosyncratic risk can sizably influence aggregate dynamics. There are only a few works that take account of the distribution of banks, they however do not take into account the aggregate effects of monetary policy transmission and the redistributive implications on banks. However, the existence of a distribution that can well characterise banks in terms of their assets, loans, marginal propensity to lend and other measures of banks' performance becomes relevant when studying the interactions between idiosyncratic uncertainty and aggregate uncertainty, because the implications from the theory for a wide range of macroeconomic questions can be changed by retaining the presence of ex-post heterogeneity.

In most of the works presented, amongst banks' balance sheet variables deposits are subject to considerable idiosyncratic risk relating to the possibility of a deposit withdrawal. In this regard, since banks cannot fully insure against the idiosyncratic shock then markets become incomplete. In models with incomplete markets, frictions in the financial sectors that affect the regulatory capacity constraint and size constraint of the banks play a crucial role for the macroeconomic and monetary policy transmission. The message from the theory is of course powerful in the sense that these frictions help to non-linearly amplify and propagate the shocks having the ability to explain to a greater extent cyclical macroeconomic fluctuations and, therefore banks need to take into consideration all the constraints they are subject to regardless whether banks are heterogeneous or not. However, one of the distinctive feature of heterogeneous banks models is in their nature of shocking directly and unexpectedly this regulatory capacity constraint, in this regard defined as a liquidity requirement or capital requirement that limits the ability of banks to expand their capital or to raise new deposits. This constraint leads banks to undertake different decisions, depending on their position in the asset distribution given the constraints they face and their inability to insure against the risk.

While discussing the details of different models, I draw emphasis on monetary shocks in the wake of the negative interest rate policies in heterogeneously affecting banks' decisions depending on their funding structure, since the empirical evidence has been proven elusive with some studies finding accommodative effects to negative interest rate changes, while other studies find contractionary effects. Therefore, I stress the relevance of aggregate uncertainty and idiosyncratic uncertainty in an environment in which the individual dynamics of the macroeconomic and financial variables do not coincide with the aggregate dynamics as stated in [Aiyagari \(1994\)](#), a shortcoming that is taken as a given in representative bank models and that, it is a non trivial task in disentangling in any heterogeneous agents models. Adding aggregate uncertainty to the model means adding computational complexity when the economic model already features many sectors and constraints that might render the problem intractable. Computational complexity arises when aggregate shocks are in place because the aggregate states become function of the joint distribution of the individual states representing the law of motion of the full total assets distribution. This aggregate distribution then becomes a relevant object that agents need to take into account in their optimal programme since it varies with monetary shocks. From a numerical standpoint, this aggregate distribution is high-dimensional and becomes computationally intensive when the state space size increases, especially with the introduction of more than one endogenous state variable.

A remarkable result able to tackle this issue was given by [Krusell and Smith \(1998\)](#) who demonstrated for the case of a technology shock in an incomplete market economy that in order to forecast all relevant future prices, interest rates and quantities, that are functions of the aggregate distribution, what matters is the mean of the distribution and the behavior of the individuals who are most wealthy in the economy since their policy functions are almost linear and their marginal propensity to save is independent from total wealth. Therefore, the infinite dimensional distribution can be approximated by a finite number of moments, a result that they baptise *approximate aggregation*. The main paradigm of the approximate aggregation result is a deviation from the full rational expectation assumption underlying many macroeconomic models. The individuals in the economy do not need to know anymore the underlying functions governing these prices and quantities, but the mean is sufficient to be able to forecast them, in addition to that, individuals must have the same constant marginal propensities to consume in order for aggregation to be possible.

However, despite being a compelling method the set of models for which it can be applied remains finite, and as soon as economic models incorporate more interesting features, then this approach may no longer be accurate. In this regard, instances in which approximate aggregation may fail are the presence of more than one inequality constraints, non linear endogenous state variables, discrete choices, non convexities in the budget constraint, linear and non linear aggregate shocks. Especially under linear and non linear aggregate shocks, the law motion over the aggregate states enters in the individual decision problem. In these cases, the approximate

aggregation problem arises because the state variable is high-dimensional and therefore it has to be approximated by a low dimensional state vector. However, a low dimensional state vector entailing only one moment does not necessarily imply an accurate characterization of the exact model solution and more moments are needed, taking caution of the fact that the solution is non linear not only in the individual states but also aggregate states of the economy. The reason for not being accurate is that agents may forget about the redistributive effects of aggregate shocks if one were to look at the mean only, therefore they need to take into account of additional effects in their decision rules captured by higher order moments.

For this reason recent computational techniques have been proposed to deal with aggregate uncertainty encompassing a wide range of solutions and methodologies, that cover parameterization of the distribution with projection methods as in [Algan et al. \(2008\)](#), [den Haan and Rendahl \(2010\)](#), a combination of perturbation and projection methods as in [Reiter \(2009\)](#), approximation by local methods through impulse responses as in [Boppart et al. \(2018\)](#), perturbation methods through finite difference [Ahn et al. \(2018\)](#), parameterization of the distribution and the use of Bayesian techniques as in [Winberry \(2018\)](#), machine learning methods as in [Fernández-Villaverde et al. \(2023\)](#), projection for square integrable random variables called polynomial chaos expansions [Pröhl \(2025\)](#), tensor grid methods and fixed copula approximation as in [Bayer and Luetticke \(2020\)](#), methods based on sequence space Jacobians and linearizations of the impulse response functions in [Auclert et al. \(2021\)](#), a combination of finite difference method and Newton algorithm as developed in [González et al. \(2021\)](#). This literature however deserves a concise separate treatment that will not be addressed in this paper. However, naturally these techniques lend to any model that aims at including bank heterogeneity in a similar style.

In estimating the heterogeneous banks' responses and the rebalancing of banks' assets following a monetary change for US commercial banks, I find a 10.9 percent reduction in illiquid assets (loans) growth for banks with more liquid balance sheets pre-financial crises, whereas post-financial crisis the effect is 0.8 percent smaller, when deposits are not accounted for. I find an important role for deposits since the impact on illiquid assets growth decreases to 2.1% and 1.5% pre and post-financial crisis. The results are reported in Section 6. These findings are line with previous empirical studies on Euro Area commercial banks that finds restrictive effects of expansionary monetary changes documented in Section 4.

Outline. The paper is organised as follows. Section 2 documents banking facts on the distribution of banks and the bank heterogeneity presents in the banking system. Section 3 presents empirical evidence on heterogeneous banks. Section 4 reviews the evidence on the heterogeneous transmission of monetary policy to banks and emphasise the important role of macroeconomic models to account for the micro level heterogeneity of banks. Section 5 reviews and examines the recent modelling frameworks of heterogeneous banks models by discussing their assumptions and implications. Section 6 validates the heterogeneity in the banking sector by performing the empirical analysis, presenting the estimation method, estimation results and

highlighting the empirical estimates implications for banks' liquid and illiquid assets. Section 7 provides suggestions for further research questions to model bank heterogeneity with stochastic volatility. Section 8 concludes.

2 Banking Facts

The banking sector has been documented as being heterogeneous along different dimensions as well as having strong concentration at the top of the asset distribution. [Boyd and Gertler \(1994\)](#), [Ennis \(2001\)](#), [Ennis and Malek \(2005\)](#) belong to the first papers in the literature documenting bank heterogeneity in their assets. They document a significant shift in the mass of US banks to the right of the asset size distribution, indicating a tendency towards a major presence of large banks and a significant reduction of small banks. As a consequence, asset concentration is now in the hand of a few large banks generating the skewness of the distribution to the right. Asset size is not the only dimension through which to understand bank heterogeneity. The cited papers have also pointed towards a strong presence of heterogeneity in banks' loans, dividends, income and the variance of banks' returns in assets and size.

Since one of the main reasons for the changes in the size distribution of banks that generated a more heterogeneous banking system has been the increased regulation in the financial system, I focus on the bank heterogeneity that arises along the capital and liquidity requirements that banks have to face. Big banks are identified as those ten banks at the top of the asset distribution, whilst all the others are considered small banks. Since 2016, the top ten banks in the US were classified as Global SIFIs. [Corbae and D'Erasmus \(2021\)](#) document three important facts presented below.

Fact 1 *Large banks have lower risk weighted capital ratios than small banks.*

The authors separate between the top ten banks and small banks during the period 1996-2016. In the US the risk weighted capital ratio is recognised as Tier I capital requirement in Basel III regulation. Regardless of the time, for both types of banks risk weighted capital asset ratios defined as, $\frac{\text{loans} + \text{net asset-deposits}}{\text{loans}}$, are above the minimum requirement, and small banks hold larger capital assets ratios than large banking institutions. Since the risk weighted capital ratio represents an average of capital across banks and over time, it might hide those banks that are at the constraint and/or violate the constraint. Moreover, the regulation recommends a requirement no lower than 6% to be defined as Well Capitalized Bank. Since both types of banks are found above this requirement, this implies that all banks have a precautionary motive driving their decisions.

Banks have a precautionary motive and the reason is that even if banks are not capital constrained, they hold large quantities of reserve ([Ennis and Wolman, 2015](#)). Recent research conducted on a more general measure of bank liquidity documents an increase in liquidity in the bottom percentile of the liquid assets distribution post-financial crisis ([Gigante, 2026a](#)). This

rise was six times larger than the rise in the top percentiles of banks' liquid assets distribution which lends support the view of a remarkable precautionary liquidity motive in banks' balance sheet.

Fact 2 *Capital buffers are smaller for large banks compared to small banks.*

A comparison of the distribution of risk weighted capital ratio in 2005 and 2010 reveals that in 2005 the risk weighted capital ratios were lower for large banks with the majority of large banks concentrated at the 6% to 10% of the risk weighted capital ratios, in contrast to small banks who tended to hold higher risk weighted capital ratios (Corbae and D'Erasmus, 2021). It is worth to emphasise that in 2005 the minimum requirement for capital ratio was 4%. Following the financial crisis 2008-09, regulation imposed a higher minimum requirement no lower than 6% that in conjunction with equity injection policies and government intervention policies shifted the distribution of banks' capital ratios to the right. As a consequence of these facts, many small banks fell below the minimum capital ratio requirement, some of them merged with other banks and others did not survive. Regarding large banks, the largest fraction of banks, approximately 30% became concentrated at around 12% of the risk weighted capital ratio, whilst other large banks either failed or merged (Corbae and D'Erasmus, 2021).

Fact 3 *Deposits and loans are procyclical. The volatility of deposits is decreasing in size.*

The lower variance of deposits of large banks than the variance of small banks explains the reason for the smaller precautionary buffer of liquidity from large banks (Corbae and D'Erasmus, 2021). Fact 3 also indicates the better ability of large banks to diversify their portfolio. Whilst the number of banks over the selected time period has been decreasing over time, the market shares over deposits of the largest banks have been increasing making the industry more concentrated as documented in Corbae and D'Erasmus (2021).

Fact 4 *Banks are heterogeneous in leverage which is higher for large banks.*

Whilst heterogeneity is present along different dimensions of banks' size, leverage is also documented to be one of the pervasive elements driving heterogeneity at bank level (Coimbra and Rey, 2024).³ As leverage increases with size, the level of leverage also becomes more sensitive to the cycle for large banks.

The need for a theory of bank size heterogeneity has historical roots and was originally advanced in a work by Janicki and Prescott (2006) who analysed the size distribution of banks from 1960 to 2005. Their work emphasises the relevance of the Pareto distribution in fitting the upper right tail of the distribution of bank size and highlights that growth is independent of bank size before the deregulation period 1960-1970, but it stops afterwards where the growth of large banks moved faster than small banks by employing Gibrat's Law.⁴ The authors consider

³Leverage is defined as the ratio between assets and equity at book value.

⁴Gibrat's Law also known Gibrat's rule of proportionate growth was pioneered by Robert Gibrat in 1931. It states that the proportional rate of growth is independent from the absolute size of an entity. While applied in the context of firms, Gibrat's Law is also applied to the size of cities. Since Gibrat's Law focusses on the whole distribution, it gives rise to a log-normal distribution.

four measures of bank size: assets, deposits, loans and the number of employees.

Fact 5 *The banking sector is characterised by many small banks and few large banks.*

The skewed distribution of the US banking sector was already demonstrated in 2005 by using the log of assets and documenting a shift to the right of the distribution and a higher fraction of assets held by large banks. In this year and subsequents, the distribution cannot be approximated by a lognormal distribution but the Pareto distribution performs a better task with the right tail of the distribution corresponding to Zipf's Law (Janicki and Prescott, 2006).⁵ Whilst the authors test this prediction by estimating the slope of the rank with a linear model using maximum likelihood, new methodology have been proposed in the literature in the recent years. Zipf's Law is shown to not capturing the accurate banks' size of large banks.

Fact 6 *Gibrat's Law does not apply to the banks after the deregulation period in the 1980s.*

The size of banks is shown to be independent of bank size, thus Fact 6 applies. Big banks become much bigger, and banks with high leverage will become more leveraged as we have seen prior and post the financial crisis. In terms of dynamics, the number of banks have been falling over time and despite a significant fraction of banks merged, the number of medium size banks remains larger than the number of large banks (Janicki and Prescott, 2006).

The banking facts documented in this section have been evolving over time and the key observation highlights remarkable heterogeneity in the banking sector that since 1960 has become even more heterogeneous. Banks have become more diverse in size. A more systematic consideration of capturing the dynamics of the banking sector in a banking industry is clearly required and this must be within a context of incomplete markets. Thus, the remainder of this paper will present the first contributions in the literature of heterogeneous banks with an overview of each model and discussion of their pros and cons.

3 Heterogeneous Banks Empirics

Empirical evidence now points out the presence of a considerable heterogeneity amongst banks. However, in order to potentially detect bank level characteristics such as the marginal propensity to lend at a macroeconomic level the researcher has to deal with the endogeneity of economic conditions and of monetary policy which often comes at the cost of making further assumptions. I will postpone the discussion of the heterogeneous transmission of monetary policy based on banks' fundamentals at a time of unconventional policies such as negative rates in Section 4 and here I focus on the micro evidence that calls for heterogeneous marginal propensity to lend.

In trying to estimate the marginal propensity to lend, Agarwal et al. (2018) run a regression discontinuity design for banks in the United States and estimate the slope of the marginal

⁵Zipf's Law is found to be associated with the upper tail of a distribution, which can be approximated with a log-normal distribution and a Pareto distribution. Zipf's Law states that in any set of values sorted in decreasing order, any additional value is approximately inversely proportional to the number of values. The law was coined by George Kingsley Zipf and usually applied to frequency distributions.

profits curve using the marginal profits curve and the average profits. They then obtain the marginal propensity to lend by parameterizing the marginal profit curve with a linear functional form. The slope of the marginal profit curve is used since it captures the impact of frictions in the relation between bank and borrower on the passthrough of credit expansion policies. The procedure requires the estimation of two moments: the marginal profits through a regression discontinuity design and the average profits observable from the data. These two moments can then be used to identify two parameters for the linear marginal profit curve. They find that the marginal propensity to lend depends on a few sufficient statistics and importantly it is negatively correlated across borrowers.

The approach based on sufficient statistics is based on the argument that only few primitives in a model have to be identified, thereby combining features of the structural approach that calibrates and estimates parameters and primitives of fully specified economic models and reduced-form approach that acknowledges the difficulty of identifying all parameters in a model. Therefore, this approach has pointed towards identifying and estimating causal and statistical relationships. The derived sufficient statistics are therefore estimated formulas for policies that depend on high level elasticities instead of primitives.⁶ In the study the authors estimate two sufficient statistics: marginal propensity to borrow and the marginal propensity to lend. Both of them resemble features of the credit market whilst at the same time neglecting some assumptions of the economic model. However, the identification problem of monetary shock is tackled by assuming that the response of monetary policy occurs optimally after any change in the cost of funds and assuming that the measurement of any bank's incentives can be detected.

I take this result since any change across banks in the marginal propensity to lend, banks' assets, capital and liquidity levels affect the bank lending channel of monetary policy and further study in these regards are Kashyap and Stein (1995), Jiménez et al. (2012), Jiménez et al. (2014), Dell'Ariccia et al. (2017) and Acharya et al. (2020) who find that a liquidity expansion by central banks has not been translated into a lower loan rate for riskier banks. This evidence points not only towards the relevance of significant heterogeneity but also about the need for frictions in the relationship between banks and borrowers on both the household side and the firm side of the market.

4 Micro Heterogeneity in Banks' Responses to Low and Negative Interest Rate Policy Disturbances

In this section I outline the importance of aggregate disturbances such as a monetary shock in delivering a different response by banks calling for a heterogeneous transmission of monetary policy across banks. A long empirical literature has investigated the consequences of conventional

⁶For early contributions on the sufficient statistics as well as a survey of the literature see Chetty (2009) and Chetty and Saez (2010).

monetary policy for the bank lending channel. These studies have addressed bank heterogeneity in the supply of credits coming from the bank's size, the holding of liquid assets and bank equity. Examples include [Kashyap and Stein \(1995\)](#), [Kashyap and Stein \(2000\)](#), [Kishan and Opiela \(2000\)](#), [Gambacorta \(2008\)](#), [Jiménez et al. \(2012\)](#), [Gomez et al. \(2021\)](#). The key point is that these works highlight that banks with more illiquid assets and predominantly located in the 95 percentile of the bank size distribution expand their loans supply by more. Moreover, banks with more liquid assets and with a solid capitalisation are more likely to adjust their lending rate after any change in the nominal interest rate.

Here, I focus on negative interest rate policies since these policies were not adopted by all central banks and because they are recent policies embedded in the monetary authority toolkit. The Denmark Nationalbank was the first monetary authority to implement negative rates on the deposit facility in 2012 after the financial crisis, followed by the Swedish Riskbank, the European Central Bank, Swiss National Bank and the the Bank of Japan. These central banks successfully implemented this unconventional monetary policy to mitigate the financial distress in the market caused by the global financial crisis, all other central banks relied on enormous amount of asset market purchases programmes whilst maintaining a low nominal interest rate. Negative interest rates have been a questionable policy because they consist of cutting in a negative region the deposit facility rate which is the interest rate banks pay to hold reserves at the central banks, and subsequently pass this fall to the interbank market rate. However, the fall in the policy rate cannot pass on the deposit rate that commercial banks pay to depositors, which remains bounded at zero since the empirical fact is that the distribution of bank deposit rates is truncated at zero.⁷

I present the recent empirical literature about the effects of negative interest rates and their heterogeneous impact on banks reactions according to their balance sheet characteristics. Table 1 summarises the recent empirical studies on negative interest rates and for each of them I emphasise the main characteristics, their strengths and limitations. This summary mainly emphasises the type of deposits being used to disentangle bank heterogeneity. Furthermore, these studies point towards the relevance of the banks' funding structure for the transmission of monetary policy and support the need of a theoretical framework able to capture micro level heterogeneity that is observed in the data.

[Heider et al. \(2019\)](#) employ a difference-in-differences estimation for the Euro Area disentangling banks according to their deposit ratio in order to detect their credit supply over the negative interest rate policy. In particular, they separate banks between banks relying more on deposit funding sources and banks less reliant on deposits, the latter are then used as a counterfactual for the behavior of banks more reliant on deposits when negative interest rates are not present. They find that over the negative interest rate period, banks that are large in their deposit funding (and therefore smaller due to their lower networth) reduced the volume of

⁷For a description of the experience of negative interest rates from the four central banks in the euro area see [Bech and Malkhozov \(2016\)](#).

TABLE 1.—Studies in the Empirical Micro Heterogeneity of Banks to Low and Negative Interest Rates

Article/Paper	Type of heterogeneity	Purpose	Effects/Findings	Strengths	Limitations
Heider et al. (2019)	Banks are heterogeneous in their total deposit ratios.	They evaluate the impact of negative interest rate according to euro area banks' characteristics through a difference-in-differences estimation. The analysis is also conducted by bank size: small and large banks.	They find a larger reduction of credit supply and greater increase of funding costs by banks more reliant on deposits upon the introduction of negative interest rates.	The endogeneity of monetary policy is addressed by comparing lending behavior of banks with more and less deposits. The identification threat on the appropriateness of the control group is overcome using supervisory information and transaction data.	The presence of missed unobserved bank level heterogeneity in the treatment and control group around the parallel trend assumption.
Altavilla et al. (2018)	Bank heterogeneity is exploited at level of banks' efficiency as measured by the cost to income ratio and non performing loan to income ratio.	They focus on the impact of low and negative rates on banks' profitability measured with the return on assets and non interest income. At micro level, they conduct a panel regression with quarter fixed effect. At macro level they perform a Bayesian Vector Autoregression to investigate the impact of monetary policy to bank profitability.	Low and negative interest rate policies are beneficial with higher efficiency and lower asset quality. An extended period of low interest rates may induce a downward pressure on bank profitability.	They use a recursive identification scheme based on ordering the last interest rates. To identify the effects of monetary policy they perform an event-study by analysing asset price changes in small windows around policy announcements to detect market reactions.	The authors acknowledge the presence of distortionary effects due to excessive risk taking and non performing loans, which however would need to be concretely included in the analysis.
Arteta et al. (2016)	Case for bank heterogeneous in their retail deposits.	The authors provide facts and evidence following the implementation of negative interest rates on the countries that implemented this policy. In addition, they perform an event study-analysis around the announcement on negative rates to investigate changes in financial variables.	To compensate the lower profit margins, banks reliant more on deposits reduce less lending rates and increase fees. Negative rates exert a fall to lending rates in countries with more bank competition and short lending maturities. Bank profitability depends on the health of the economy impacting the volume of lending.	The provision of sources and implications of negative interest rates touching several noteworthy points, from the consequences of this policy to banks, to the economy and at international level.	The inclusion of bank level characteristics in the event-study over the negative interest rate window.

NOTE. This table summarises articles and papers that assessed the adoption of the negative interest rate policy in the set of heterogeneous banks.

Article/Paper	Type of heterogeneity	Purpose	Effects/Findings	Strengths	Limitations
Amzallag et al. (2019)	Bank heterogeneity is examined in terms of banks' funding profile, specifically retail overnight deposits.	The authors study the implications of negative interest rates on mortgage lending market in Italy through the lenses of a regression analyses by using granular loan level data.	Banks with higher overnight deposits ratio charge higher mortgage rates with a non significant difference for other types of banks. The impact is larger for fixed rate mortgages when the sample is extended from 6 to 12 months before and after the introduction of negative interest rates and for more constrained banks who compensate for the lost interest income. The negative effect on households due to higher lending rates are negligible suggesting the concerns of negative rates are less relevant.	The analysis conducted through a time interaction term on the main control variables, overnight deposits ratio, is isomorphic to a difference-in-differences estimation. Use of fixed effects and time-varying characteristics interacted with the time of negative interest rates.	The presence of confounding macroeconomic factors upon the introduction of negative interest rates not explicated in the analysis due to the introduction of other monetary policy measures simultaneously to negative interest rates. Financial indicators of the state of the economy and bank specific that may pose a threat to identification. The possibility to consider a double and triple interaction term.
Demiralp et al. (2021)	Bank heterogeneity is exploited at the level of retail deposits and excess liquidity holding jointly. This variation is also used as a new identification scheme.	The paper studies the heterogeneous banks' impact of negative interest rates on loans and its mechanism for euro area banks using confidential bank level data through a panel fixed effect estimation. The analysis is also conducted by bank size: small, medium and large banks.	The authors find negative interest rates being an accommodative monetary policy. The finding is brought by the excess liquidity ratio that qualifies a portfolio rebalancing mechanism to loans that is larger for banks more reliant on deposits and with more liquid balance sheet. This in turn leads to more risk being taken by this type of banks.	The authors distinguished themselves for having studied a two-side heterogeneity. Having explicated in the analysis the direct costs of negative interest rates through the cost associated to excess liquidity, namely reserves in excess of required reserves. The same variable is able to capture for the rebalancing of low yield assets to those with higher return.	Restricted measure of excess liquidity. Generalisability of results to different samples including different types of bank loans.

NOTE. See associated note at Table 1.

Article/Paper	Type of heterogeneity	Purpose	Effect/Findings	Strengths	Limitations
Claessens et al. (2018)	Banks heterogeneity with different deposit to liability ratios.	Panel regression analysis on euro area banks to analyse the impact of low interest rates to banks' net interest margin.	The study indicates large net interest margins for small banks and negligible differences in large and small banks' profitability.	The correlation of low interest rates with economic conditions controlled by holding constant other factors.	The authors recognise limits to appropriateness of lags and nonlinearities in interest rate changes and bank profitability.
Bottero et al. (2019)	Banks are heterogeneous in their ex-ante and short term interbank market positions and securities to assets ratio.	They study the impact of negative policy rates on credit supply growth for euro area and Italian banks through credit register data matched with firm and bank level balance sheet data by performing a difference-in-differences estimation.	The empirical findings highlight a positive impact on credit supply for banks with more ex-ante and short term interbank market position to riskier and smaller with a simultaneous fall in loan rates and raise of fees in bank deposits that save bank profitability. Negative interest rates expand economic activity and have stronger effect for banks with <i>higher</i> liquid to total assets ratio.	The findings rely on a rebalancing mechanism from liquid assets to illiquid assets and not a retail deposit channel.	Possible bias in the ordinary least square estimates of the difference-in-differences. Inclusion of controls for the yield curve. Limit to the generalisability of the results to a sample of banks heterogeneous in <i>ex-post</i> interbank market positions and liquid balance sheets.
Boungou (2019)	Bank heterogeneity exploited according to the liquidity to assets ratio and bank size.	They study the effect of negative interest rates on a bank profitability measured by net interest margins and profits and on risk taking measured with non-performing loans, provisions and z-score on bank's solvency for a set of 28 European countries using a dynamic panel estimation.	Negative interest rates induced a fall on banks' margins larger for larger banks, taking less risk in contrast to the theoretical guidance. Moreover, they estimate a 0.47% reduction in banks' margins to 1% decay of rates in negative territory. The effect is higher for banks with low liquidity who also experience a fall in profitability.	The endogeneity bias due to reverse causality and omitted variable bias is overcome by using a lagged dependent variable and dynamic panel estimation through a system of generalised method of moments.	The impact of negative interest rates is not directly measured by banks' size.

NOTE. See associated note at Table 1.

Article/Paper	Type of heterogeneity	Purpose	Effect/Findings	Strengths	Limitations
Boungou and Mavusi (2023)	Banks are heterogeneous in terms of their total deposits to assets ratio, assets and equity capitalisation.	The assessment of negative interest rates on banks' lending margins and bank lending using an unbalanced panel of euro area banks with the adoption of a difference-in-differences estimation.	The findings reveal that small banks more dependent on deposits and less capitalised banks suffered the most from the fall in bank lending margin.	The estimation is further conducted in samples divided between large and small banks.	Mispecification of the control group if banks unaffected by negative interest rate policy becomes affected by the policy in the estimation window. Non random assignment.
Eisenschmidt and Smets (2019)	Bank heterogeneity captured in total deposits.	Using confidential bank level data for euro area banks, the authors assess the negative interest rates policy to lending rate and the volume of loans.	Bank more reliant on deposits are less willing to pass the rate to loan rate. There is no evidence of contractionary effects of this policy.	The fixed effect panel estimation is able to capture the lack of a change in passthrough following negative interest rate policy.	Restricted estimation that can consider other control variables to measure the passthrough at a bank level.
Altavilla et al. (2021)	Note for bank level heterogeneity considered in the strength of their balance sheet.	They study bank credit supply and demand over the unconventional policy of negative interest rates in the euro area through a difference-in-differences. Excess liquidity and bond holdings are used as proxy for exposure of banks to negative interest rates.	Findings reveal an increase in loan supply by banks more affected by the policy due to easing in negative territory of the policy rate.	Results are robust to considering a pooled OLS, time fixed-effects and country time-fixed effects.	Estimation can be conducted according to bank size.
Lopez et al. (2020)	Bank heterogeneity exploited in the bank deposit ratio and bank size by assets.	The paper assesses negative interest rate policy on banks' profitability for a sample of Japanese and euro area banks by conducting a difference-in-differences estimation.	The authors find heterogeneous effect of negative rates with low deposit banks experiencing an increase in the non interest income, lending activity and deposit funding. Banks reliant more on deposits experience large falls in net interest income.	The authors address the endogeneity of monetary policy by controlling for local conditions: unemployment, GDP growth, output gap and inflation.	Results not particularly significant in the sample of large and small banks. Treated and control groups characteristics after the negative interest rate policy.

NOTE. See associated note at Table 1.

Article/Paper	Type of heterogeneity	Purpose	Effect/Findings	Strengths	Limitations
Altavilla, Laeven and Peydró (2020)	Bank heterogeneity in the level of capitalisation.	They measure the monetary policy effect of negative interest rates and macroprudential policy using loan and bank level data for the euro area through a panel fixed effect.	Less capitalised banks respond more strongly and positively on lending following complementarity in both policies, while enhancing risk taking.	Results conducted for households and firms separately from banks. Double and triple interaction terms, necessary to detect complementarity between monetary and macroprudential policies.	Correlation of loans with monetary policy that bias the identification due to bank level characteristics that have not been taken into account in the estimation such as bank leverage. Usage of few controls.
Bittner et al. (2020)	Bank are heterogeneous in their total deposits structure.	The authors analyse the passthrough of the policy rate in a negative territory to banks' cost of funding for euro area countries through a difference-in-differences estimation using confidential credit registered, bank and firm level data for Germany and Portugal.	While in Germany banks more reliant on deposits increase lending to risky firms that are financially constrained, in Portugal less capitalised banks expand their lending to risky and safe firms without any impact from variations of banks' funding structure to loans. Negative nominal rates do not transmit to lower deposit rates.	The cross-country comparison with the rest of countries is modelled as convex combination of the banking system in Germany and Portugal.	Results depend on country-type that drives them, external validity may be impaired. Identification problems present by the endogeneity of monetary policy due to lack of bank time fixed effect in the estimation.
Ampudia and Van den Heuvel (2022)	Bank heterogeneity in deposits intensive funding to total liabilities ratio.	Assessment of exposure of banks to interest rate risk in a negative interest rate environment. Event-study analysis to examine monetary policy announcement on bank stock returns due to change in interest rates in the euro area across banks.	Decay of the net interest rate margins of high deposit banks and absent change for low deposit banks. The triple difference-in-differences reveal a fall of stock price markedly higher for high than medium and low deposit banks.	Estimation conducted by bank size: high, medium and low. Cross sectional variation captured by the deviation of bank size from its time specific mean.	Reverse causality that drives interest rate and swaps at different maturity. No consideration for within bank fixed effect.

NOTE. See associated note at Table 1.

Article/Paper	Type of heterogeneity	Purpose	Effect/Findings	Strengths	Limitations
Altavilla et al. (2022)	Banks are heterogeneous in their level of soundness.	The authors study the interest rate passthrough with the policy rate in negative territory for euro area banks using confidential data through local projections.	Sound and safe banks transmit negative policy rates to corporate depositors without experiencing a fall in credit, owing to the greater market power these banks.	Results robust to time varying bank characteristics, bank and country fixed effects. Firms linked to banks with high passthrough reveal that they rebalance their assets from liquid assets to loans.	Inclusion in the estimation of market power for sound banks to be examined through cross-corporate-product at bank level.
Madaschi and Pablos Nuevo (2017)	No explicable bank level heterogeneity considered.	The authors examine banks' profitability and the transmission of monetary policy focussing on banks' lending, deposit rates and funding in Sweden and Denmark. Bank lending and deposit passthrough analysis is conducted through an error correction model.	Findings highlight that banks' profitability and credit have continued to improve maintaining the transmission of monetary policy to lending rates for these countries, who squeezed interest rate margin.	The authors provide a collection of stylised facts for the countries under study over the negative rates.	The inclusion of a bank level dimension in the analysis for the selected countries and the support of descriptive statistics.
Arce, Garcia-Posada, Mayordomo and Ongena (2020)	Bank heterogeneity in terms of the level of capitalisation based on the capital ratio.	The paper assesses the long term impact of negative rates on bank lending and lending rates by matching bank level with credit register and firm balance sheet data in Spain. By conducting a difference-in-differences they address the size dimensionality through interaction terms with low and high capital banks.	Their results corroborate the view that low capitalised banks and more affected by the policy reduce more credit supply to firm. The effect is transitory and only materialise after the negative rate policy in 2018-2019.	The identification task is carried out by estimating a probit model able to capture the probability that a bank is affected negatively by negative rates. They compare lending decisions of multiple banks to a firm within period. Triple interaction term: 'policy $\times$ size $\times$ time'.	The possibility to control for mergers. Similarly, the possibility to perform a within-generalisation of the results for the same sample but for high and low capitalised banks separately without the bank size interaction term.

NOTE. See associated note at Table 1.

credit supply compared to banks relying less on deposits and the few available loans were given to riskier borrowers. In particular, aggregate bank lending reduced by 13% after an increase in the standard deviation of a bank's deposit ratio corresponding to an increase of 14.76 percentage points; whilst, the volatility in bank return on assets increased by 16% brought by an increase in the volatility in bank's deposit ratio of 9.45 percentage points, meaning that during negative interest rate policies banks with more deposits increased their loans to riskier firms. There is a potential issue of endogeneity of monetary policy due to the economic conditions that is controlled for by considering the difference in the lending behavior between high and low deposit banks, otherwise the results are likely to be biased.

Altavilla et al. (2018) adopt a panel regression for Euro Area banks obtaining beneficial effects for banks more efficient in terms of having a low cost to income ratio and less quality assets benefit mostly from negative policy rates. They underline the importance of endogeneity of monetary policy with aggregate economic conditions by providing significant results on negative banks' profitability in the short run if the right control variables are not taken into account to address the endogeneity problem. The panel data on banks is then combined with a dynamic multivariate model in order to conduct a high frequency event study for the aggregate reaction of market participants to expected change in banks' profitability given the adoption of negative rates. The event study analysis consists of using intra-daily data on asset prices in a two-day window around policy announcements capturing information that was available to the market during the day, thereby identifying the endogenous response of changes in monetary policy and changes in stock prices as well as signal coming from macroeconomic news that can cause the joint response of changes in monetary policy and stock prices.⁸

Therefore, a narrow window is chosen as an identification strategy in such a way that there is no influence of assets prices changes such as stock prices and credit default swap and macroeconomic news on monetary policy decision. Hence the causality impact from the policy to asset price changes is properly accounted. In this way monetary policy changes are unanticipated for market participants and therefore assets prices will react to this monetary news only and to no other factors. The authors find a positive impact on banks' market valuations with an increase in banks' stock price of 1.5% approximately. However, if negative rates are set to remain in the medium long run then bank profitability can be seriously impaired.

In line with this reasoning is also Arteta et al. (2016), who point that the deterioration of bank profitability may come not because of negative rates but because of a simultaneous interaction of the health of the economic system and monetary policy actions.

Focussing on banks classified in terms of their overnight deposit ratio to total assets is Amzallag et al. (2019) who study the implications of negative rates on mortgage rates for the Italian banking sector. By running a difference-in-differences estimation for Euro Area banks, they find that banks high in overnight deposits set higher rate on new fixed rate mortgages in

⁸For early event study analysis see Bernanke and Kuttner (2005), Gürkaynak et al. (2005).

order to maintain their profitability. In addition to it, they find that the overall aggregate loss for households implied by higher loan rates in order to access credit is small, concluding that negative rates are not inefficient as a tool for monetary policy.

Banks categorised between banks relying more on deposits holding and banks with less deposits in their funding structure are analysed in [Demiralp et al. \(2021\)](#). They disentangle heterogeneous banks not only through the holding of deposits, but also of their excess liquidity jointly. They run a panel fixed effect estimation for Euro Area banks and study negative interest rate policies through the lens of a rebalancing of portfolio variables. In particular, they uncover a rebalancing of assets from liquid reserves to illiquid loans for banks relying more on deposit holdings. As a result and in contrast to [Heider et al. \(2019\)](#), they find a positive effect on loan supply by banks more reliant on deposits.⁹ The volume of adjustment depends on banks' excess liquidity, those reserves above the minimum reserve requirement, since determines the charge born by banks on the deposit facility rate. The larger the excess liquidity banks hold and the higher the leverage ratio, the larger the adjustment in the balance sheet they undertake. However, in the robustness estimation these liquid assets are broadly defined since they do not consist of only reserves held at the central banks, but include the sum of wholesale funding, government bonds, debt securities and equity. The identification strategy is carried out by including an interacting term for the excess liquidity ratio with a dummy variable for the negative interest rate window.

Along slightly different lines, [Bottero et al. \(2019\)](#) also exploit the impact of negative rates on banks' profitability and bank risk taking over the period 2013-15. However, they do it by analysing data that capture the supply of loans such as credit registered data and inspect the consequences of negative rates through a portfolio rebalancing from liquid assets. It consists of two measures: interbank market positions (interbank market loans minus interbank market deposits) and the ratio of securities over total assets since negative rates impacting the yield curve had an effect on all maturities, towards illiquid assets or credit supply. They use a difference-in-differences estimation including a fixed effect term for the identification of the supply of loans by banks. They find a reduction in lending rates of 4.4% in consequence to a change of one standard deviation in liquidity positions and an increase in loan supply by 1.4% following the same change in liquidity through a portfolio rebalancing mechanism of liquidity to illiquid assets for banks mostly affected by negative rates policy; in contrast to [Heider et al. \(2019\)](#) who estimate a reduction in loan supply over the negative rate period through a deposit channel by which banks with large deposits holdings reduced their credit.

However, both of these empirical evidence ([Heider et al. \(2019\)](#) and [Bottero et al. \(2019\)](#)) agree that the credit given to the economy in the period under study was given to riskier and

⁹Analysing the experience of Japan, [Inoue et al. \(2019\)](#) find a fall in bank loans from banks relying excessively on deposits by exploiting a difference-in-differences for a panel of firm-bank data following the negative interest rate policy. The aftermath is consistent with the findings of [Heider et al. \(2019\)](#) for a reduction in banks' loan supply.

smaller firms.¹⁰ Bottero et al. (2019) results stand also in contrast to studies of the bank lending channel under standard conventional monetary policy, previously analysed by Kashyap and Stein(1995, 2000), Jiménez et al. (2012) who find that the largest increase in loan supply was due to banks with low liquid to total assets ratio, rather than high liquid to total asset ratio as Bottero et al. (2019) evidence. The rebalancing in asset positions from liquid to illiquid assets in Bottero et al. (2019) findings occurs because the reduction in negative territory of the policy rate led to a downward shift and a flattening of the yield curve on safe and riskless assets enlarging the difference between the yields on liquid assets and risky illiquid assets. In terms of profitability, Bottero et al. (2019) observe a preservation of banks' profits that occurs because banks raise fees on retail deposits.

On the other hand, Bounbou (2019) uses an unbalanced panel for Euro Area banks over the period 2011-17 in order to inspect the effect of negative rates on risk taking and profitability. They tackle the endogeneity of monetary policy, bank profitability and bank risk taking by applying Arellano and Bond (1991) generalised method of moments. Profitability is measured with either net interest income or the return on assets, the former results in a drop of 0.43 percentage point after a one percentage change in interest rate in negative region by the central bank whereas the latter does not deliver any impact on banks' profits. As a proxy for bank risk taking they use non performing loans, and they find a statistically significant 2.4% decrease in banks' risk taking in contrast to Heider et al. (2019).

Similarly to Bottero et al. (2019), Bounbou (2019) assesses the impact of negative rates in bank heterogeneity disentangled between banks with a large liquid to assets ratio and small liquidity ratio and banks small and large in terms of their size captured as total assets. They identify small banks as those banks below the median of the distribution of the logarithm of total assets and liquidity ratio. They find a large reduction in banks' interest rate margin, larger for banks holding few liquid assets with a drop of 47% compared to banks with more liquid assets that experienced a drop of 37%. With regard to the size of banks, the reduction in the interest rate margin is greater for banks of large dimension than small banks, with a decrease of 92% and 37%, respectively. Bounbou and Mawusi (2023) run a difference-in-differences estimation for a set of European countries to investigate the impact of negative rates on banks' lending margins. The empirical findings reveal a fall of bank lending margins greater for smaller and less capitalised banks as well as banks more dependent on deposits. Contrastingly to Heider et al. (2019) and in line with Demiralp et al. (2021) and Bottero et al. (2019), they find an increase in lending by banks more reliant on deposits.

The key point in the portfolio rebalancing mechanism is that if banks are indeed rebalancing their portfolio from liquid assets through loans, then this reallocation must come through frictions that a theoretical framework with heterogeneous banks needs to represent. The mechanism from liquid assets to illiquid assets become important because liquid assets increased from 0.6 to

¹⁰Lending also contracted in France due to high deposits banks facing high costs of funding. These banks rebalanced their loan portfolio from large and safe firms towards small and risky firms (Duquerroy et al., 2020).

5.6 percent for Euro Area banks, as estimated in [Eisenshmidt and Smets \(2019\)](#). This study by [Eisenshmidt and Smets \(2019\)](#) uses descriptive statistics and also tests the bank lending channel in the negative rate regime. Interestingly, they do not find any change in the loan supply of large deposit holdings banks and their interest rate on loans following the introduction of negative rates, whereas [Altavilla et al. \(2021\)](#) find an increase in loan supply attributing the results to heterogeneity in risk exposure. While the difference-in-differences estimation does not control for bank size, results are robust to different types of estimation that consider pooled OLS, panel fixed-effects and country time-fixed effects. A higher level of excess liquidity is used as proxy for banks most affected by the negative interest rate policy. The analysis however, does not consider the estimation according to bank size which can neatly pin down the impact of this policy on bank credit supply.

Similarly to [Heider et al. \(2019\)](#), [Lopez et al. \(2020\)](#) also distinguish banks between high and low deposit banks and in terms of their size with respect to the holding of assets. However, in contrast to [Heider et al. \(2019\)](#), they find an increase in bank lending for banks large in their deposit holdings but small in their size, whilst banks with few deposits experienced a decline in lending and therefore a decline in their interest income. Moreover, the shift from liquid assets to illiquid assets for small banks with high deposits was the largest in comparison to the decline in liquid assets from banks relying more on deposits.

The reason may be that [Lopez et al. \(2020\)](#) use negative rates as an instrument proxied for local conditions relevant to the conventional monetary policy such as: unemployment, output gap, inflation. As for bank profitability, they address the question through a panel OLS regression in which the slope coefficient represented the effect of negative nominal interest rates on banks' net interest income. They find that negative rates induced a deterioration of banks' profitability, albeit small, and the effect was relatively larger for banks with more deposit holdings. Additionally, the decline in net interest rate income was accompanied by an increase in non interest income of similar size.

Over the unconventional monetary policy period the behavior of heterogeneous credit supply is also examined by [Altavilla, Laeven and Peydró \(2020\)](#) who consider the joint interaction of monetary policy and macroprudential policies on credit for euro area banks. By employing a panel fixed effects with multiple interaction terms, they confirm the result that accounting for heterogeneous bank funding structure is important to determine the behavior of credit supply. In particular, they find that an expansive monetary policy leads weak banks (with less capital) to lend more and to riskier firms. However, in a period of financial distress if macroprudential policies are expanded when the negative interest rate policy has already been adopted then the additional lending can be directed to the more productive riskier firms. The result suggests that the coordination between monetary and macroprudential policies can be optimal if inspected only through the increase in loan supply, but can turn out to be suboptimal if additional credit is extended to riskier firms. This indicates that further theoretical frameworks need to be

developed taking into account the heterogeneity of the banking sector and the optimal complementarity between macroprudential and monetary policies. Recent theoretical frameworks that consider the sole macroprudential policy and its effect on credit are [Lorenzoni \(2008\)](#), [Bianchi \(2011\)](#), [Farhi and Werning \(2016\)](#), [Bianchi and Mendoza \(2018\)](#), [Jeanne and Korinek \(2019\)](#) and [Malherbe \(2020\)](#).

The implications of negative rates are extensively documented in [Rogoff \(2017\)](#), [Rogoff \(2020\)](#) and [Lilley and Rogoff \(2020\)](#).¹¹ They argue the need of setting unconstrained policy rates by the monetary authority in order to mitigate the effects of a slowing down economy, proposing alternative methods on how to avoid a run into cash if negative policy rates are passed on depositors through negative deposit rates.¹²

Since the concern has been mainly on bank's profitability, several studies have further inspected the impact of negative rates. On the one hand, [Ampudia and Van den Heuvel \(2022\)](#) conduct a high frequency event study methodology to identify the policy rate surprises on bank stock prices for Euro Area banks to assess the impact on monetary policy around policy announcements. Their focus is on bank's equity, hence they evaluate how the induced bank stock prices vary according to banks balance sheet information. In particular, they run a difference-in-difference-in-differences estimation and separate banks between high and low deposit banks. The result is a significant contraction of banks' stock price of about 2% and net interest rate margins for banks heavily relying on deposits since these banks in general engage more with maturity mismatch, which confirms the hypothesis postulated by [Abadi et al. \(2023\)](#), by which further cuts below the reversal interest rate become contractionary in order to support lending. Likewise, [Claessens et al. \(2018\)](#) find a contraction in bank's profits and lending with the banks of large dimension in securities to asset ratios experiencing the more sizable drop in a low interest rate environment.

On the other hand, [Altavilla et al. \(2022\)](#) first estimate a local projection model for banks' deposit rates to changes in the deposit facility rate finding a little pass through on deposit rates. They find that sounder and healthier banks pass the negative rate to corporate depositors without experiencing a fall in deposits and credit. A possible explanation is that the aftermath relies on banks' market power for this type of banks to set corporate deposit rates. Yet, market power is not examined in the estimation. Banks' soundness is proxied with investment grade, credit default swap spreads and non performing loans. They then estimate the impact of negative rates on firms' profitability and obtain that corporates with more liquidity did not experience a drop in their profits. Rather, they increased their investment in illiquid assets in order to avoid paying the cost associated with negative rates on liquidity thereby rebalancing their assets.

¹¹The positive aspects of negative interest rates are also discussed in [Agarwal and Kimball \(2015\)](#) and [Agarwal and Kimball \(2019\)](#). For documented facts of unconventional monetary policies see [Dell'Ariccia et al. \(2017\)](#) and for a recent literature review see [Heider et al. \(2021\)](#).

¹²For the heterogeneous transmission in the interest rate passthrough, not entirely focussing on negative rates policy, see [Holton and d'Acri \(2018\)](#), [Altavilla, Canova and Ciccarelli \(2020\)](#), [Bittner et al. \(2020\)](#), among others. See [Drechsler et al. \(2017\)](#) for the deposit channel of monetary policy through interest rates.

Early studies on the effects of negative rates on banks' profitability, such as [Jobst and Lin \(2016\)](#) and [Madaschi and Pablos Nuevo \(2017\)](#) also conclude that there is no notable banks' profits deterioration and that the standard transmission mechanism of an expansionary monetary policy is still in place when negative rates are employed.

Likewise, [Arce, Garcia-Posada, Mayordomo and Ongena \(2020\)](#) study the negative interest rate policy in the context of a prolonged time 2014-2019 and its impact on bank lending and its transmission to lending rates. By disentangling banks according to their level of capital ratio, they merge bank level data with credit register and firm level data in Spain. In a first step the authors estimate the probability for banks to be adversely affected by the negative rate policy through a probit model, where this probability is linked to the profitability of banks according to the net interest income. In this step, results reveal that weaker banks with more deposits and more liquid balance sheet report adverse effects on their profitability. This information is then used in a second step for the identification of the main difference-in-differences estimation. The identification is further controlled by interacting the probability estimated in the previous step with time dummies and including firm time fixed effect thereby comparing multiple banks' lending to a firm within the same period.

The findings indicate that low-capitalised banks reduce credit supply to firms but only post negative interest rate period in 2018-2019, whilst high capitalised banks did not change their lending decisions. Furthermore, their results suggest that a prolonged period of negative rates pose concerns to bank profitability by curtailing the intermediation activity and cutting net interest margin. In contrast to previous studies, their bank size estimation is conducted by implementing several interaction terms where the dummy variable for high and low capital banks is interacted with the main control variable on whether banks are affected by negative rates. By conducting a two-dimensional heterogeneity between high and low capital banks and risky and non-risky firms, the authors conclude with a statistically significant result that low capitalised banks reduce more credit to risky firms relative to banks non affected by negative interest rates. Consistently with [Amzallag et al. \(2019\)](#) the authors find that the passthrough of the policy rate to lending rates is incomplete and low capitalised banks implemented higher lending rates to risky firms than high capital banks, thereby posing concerns for financial stability.

This class of models use the identifying assumption of banks being more and less reliant on deposits whereby small banks and banks relying more on deposits should be more affected than banks with low deposit holdings since they are more likely to make losses under negative rates.

The key take away from this section is that, the heterogeneous transmission of monetary policy hinges on the heterogeneous initial conditions in banks' composition of assets, size and their degree of liquidity. Findings on negative interest rates policies indicate that this policy remains controversial from an empirical perspective, a cut in negative territory of the policy rates may have accommodative or contractionary effects. The scant theoretical literature on negative rates to date needs to incorporate bank heterogeneity explicitly in order to provide

conclusive guidance on their distributional effects to individual banks and their impact to the macroeconomy. The vast majority of empirical studies analysed in this section reveal an overwhelmingly presence of bank heterogeneity at the level of total deposits. In periods of financial market disruptions macroprudential policy can mitigate the financial distress by affecting more banks with few illiquid assets, therefore macroeconomic models of banks need to account for this micro level heterogeneity. The studies examined in the following section have taken a step forward in this direction.

5 Heterogeneous Bank Models and their Macroeconomic Implications

In this section I discuss the current state of heterogeneous banks models in macroeconomics and I study into details their assumptions and implications, while at the same time I try to point out their virtues and restrictions.

The seminal works of [Townsend \(1979\)](#), [Bernanke and Gertler \(1989\)](#), [Carlstrom and Fuerst \(1997\)](#), [Kiyotaki and Moore \(1997\)](#), [Bernanke et al. \(1999\)](#), were the first contributions in macroeconomic models dealing with financial frictions on the side of non financial corporations and embedded as informational or agency problems. Subsequent models related financial frictions to financial corporations, examples are [Diamond and Rajan \(2001\)](#), [Gertler and Kiyotaki \(2010\)](#), [Gertler and Karadi \(2011\)](#), [Brunnermeier and Sannikov \(2014\)](#), [Cúrdia and Woodford \(2016\)](#), [He and Krishnamurthy \(2019\)](#). All of them agreed that financial frictions can have not only persistent effects but also shock amplifications that can often deliver financial volatility and instability.

In addition to the credit risk, banks have to deal with liquidity risk, the risk that in any given point in time a bank may be subject to an expected change in its liquid funds, destabilizing its balance sheets and potentially transmitting to the entire economy with a drying-up of liquidity. However, these models feature complete markets and a representative homogeneous bank or although featuring incomplete markets still deliver ex-post homogeneous banks that cannot make predictions about how the effects of either an aggregate shock and/or an idiosyncratic shock varies across the distribution of banks which sometimes may depend on the amount of liquid assets that banks hold in addition to their leverage.

Table 2 summaries the articles and papers whose models are analysed in details in the following sections. In the Table I focus the attention on those models that fall within a topical area and are tied by a common relevant factor. Furthermore, I emphasise the models in which there is a perceptible form of bank heterogeneity and I highlight their strengths and limitations. To this end, I provide a taxonomy with which I classify banks according to their type of heterogeneity.

I attribute to models either extrinsic, ex-ante and/or ex-post heterogeneity. I define extrinsic heterogeneity when a model is characterised by a perceptible form of bank heterogeneity such as two-bank, three-bank and so on, however ex-post banks can be aggregated to one representative

bank. For the models identified in this paper, I denote with ex-ante bank heterogeneity those models where banks enter the economy heterogeneous however ex-post banks are exclusively characterised by a single representative bank. In both cases heterogeneity vanishes, aggregation does not matter when models aggregate to a representative bank and we need to move to models where heterogeneity matters. Finally, ex-post bank heterogeneity is the heterogeneity where banks survive differently due to incomplete markets and are therefore characterised in the aggregate by a distribution.

Some of the models presented in the following sections still fall in the first and second category, *i.e.* featuring ex-post homogeneous bank despite having bank heterogeneity, and only few exceptions are present in the literature that however come with further model restrictions. The few exceptions have been accompanied by a growing research conducted to date that put at the centre state the important role of bank heterogeneity in theoretical and quantitative macroeconomic models.

I start by presenting the two articles on which I extensively focus, [Bianchi and Bigio \(2022\)](#) and [Corbae and D’Erasmus \(2021\)](#), that contain important features of heterogeneous banks which emphasise relevant dynamics in the banking industry. After describing these models, I restrict the framework to some features of the banking industry to present a general model structure that is simple enough to embed the aforementioned articles. I therefore adopt a top down approach.

In [Figure A.1](#) in [Appendix A](#) I provide a pedagogical taxonomy of the bank heterogeneity studies that I discuss in this section. Models are classified in macro area according to their distinctive features.

5.1 Bank Heterogeneity and Liquidity Management

Modern models of the banking system are formalizing heterogeneous banks in a framework that underlines the interaction between liquid assets and the interbank market. [Bianchi and Bigio \(2022\)](#) build a framework where interbank market frictions are able to affect the liquidity management of banks that triggers the credit channel of monetary policy. They identify a theoretical mechanism by which monetary policy can influence the bank supply of credit by affecting the trade-off resulting from high returns given by illiquid assets such as loans and the low returns of liquid assets such as reserves and government bonds. The trade-offs between these instruments are encapsulated into banks’ liquidity premia. The economy is in a real equilibrium since monetary policy acts through the central bank balance sheet and not through the presence of nominal rigidities in a New Keynesian designation. It is therefore this balance sheet that gives rise to the price level in the economy and real effects are delivered through the liquidity premia of banks.¹³

¹³See [Arce, Nuno, Thaler and Thomas \(2020\)](#) and [Piazzesi et al. \(2019\)](#) for the integration of the interbank market with nominal rigidities in a New Keynesian monetary policy. In these models heterogeneous banks are heterogeneous in their initial states, they are used to explain the model and as in [Piazzesi et al. \(2019\)](#) and [Bianchi and Bigio \(2022\)](#) they only help to create an aggregate demand for liquidity, but again macro aggregates

TABLE 2.—Studies in the Theoretical Micro and Macro Heterogeneity of Banks

Article/Paper	Purpose	Type of heterogeneity	Solution Method	Strengths	Limitations
Bank liquidity and monetary policy Bianchi and Bigio (2022)	The authors study the bank liquidity management over the financial crisis in the US through the lenses of a frictional interbank market.	Ex-ante heterogeneous banks in deposits, ex-post homogeneous bank.	Nonlinear system of two equations in two unknowns.	Homotheticity and quasi-linearity in households preferences make aggregation in the banking sector viable despite the nonlinearity of the liquidity yield function. The connection of monetary policy through bank liquidity premia.	The characterisation of the interbank market through random frictions that is amenable to direct search frictions to jointly match interbank market rate, distribution of trading banks in each session and intermediated funds. Impossibility to infer distributional monetary consequences.
De Fiore et al. (2026)	The paper studies the decline in uncollateralized funds and increase threshold accepted collateral through interbank market frictions connected to monetary policy interventions in a dynamic stochastic general equilibrium model.	Extrinsic heterogeneous two-bank model, ex-post homogeneous bank.	The model can be solved by linear solutions.	Interaction of five inequality constraints giving rise to macroeconomic policies.	The 'big family' assumption deliver an aggregate bank. Individual banks' balance sheet do not depend on distributions.
Bank distributions and banks' market power Corbae and D'Erasmo (2021)	The authors analyse the impact of regulatory and liquidity requirements to banks on banks' concentration and market industry measures.	Ex-post heterogeneous banks in deposits.	Value function iteration adapted to the industry equilibrium.	The imperfect competition à la Stackelberg with one big bank and many small banks. Entry and exit dynamics that makes the distribution of banks dependent on entrants.	The impossibility to account for mergers in the banking system within this setting. The model considers only one big bank, cannot account for a small mass of big banks.

NOTE. This table summarises articles and papers that deal with bank heterogeneity. Each model is categorised with its type of heterogeneity: extrinsic heterogeneity, ex-ante heterogeneity, ex-post heterogeneity. Models are classified in six categories.

Article/Paper	Purpose	Type of heterogeneity	Solution Method	Strengths	Limitations
Corbae and D'Erasmus (2020)	The paper assesses the impact of the Riegle-Neal Interstate Banking and Branching Act of 1994 through a reduction of the cost of expanding funding base.	Ex-post banks in deposits.	Value function iteration adapted to the industry equilibrium.	The ability of the model to match data counterpart in deposit and loan market share without targeting banks' concentration. Expansion of banks pre and post reform through the interaction between deposit process and investment.	The authors acknowledge the limit of the model to capture mergers through an endogenous mechanism and the lack of transitional dynamics.
Dempsey and Faria-e Castro (2025)	The paper analyses lending relationship representing market power on loans modelled as deep habits between banks and borrowers and banks' recapitalisation.	Ex-post banks in deposits and loan returns.	Value function iteration applied with nested golden section.	The type of competition perfect or imperfect driven by the adjustment costs in lending relationships. With lower cost the borrower keeps borrowing from the same bank.	The dynamics of loan supply abstracts from the mass of entrants that is important to endogenise as the focus is on the propagation of the shock.
Liquidity Regulations					
Gigante (2026a)	The author studies the accumulation of banks' liquid assets post-financial crisis and liquidity crisis due to deposit withdrawals.	Ex-post banks in deposits.	Fixed point iteration.	The ability of ex-ante regulations and two assets side to engender sorting by type, inequality in banks' liquid assets and constrained banks. Both of them, foster banks' precautionary behavior that tempers liquidity crises.	The second asset is exogenous hence the model cannot account for the marginal propensity to lend.

NOTE. See associated note at Table 2.

Article/Paper	Purpose	Type of heterogeneity	Solution Method	Strengths	Limitations
Gigante (2026b)	The paper examines how liquidity regulations and adjustment costs in illiquid assets determine the market structure of banks.	Ex-post heterogeneous banks in deposits.	Fixed point iteration, value function iteration with Howard improvement step and Nelder-Meade optimisation.	The model captures endogenous interactions between state dependent liquidity and leverage constraints and adjustment costs in illiquid assets whose technology generates negative assortative matching via an assignment matching game and a redistribution between banks and across assets.	The model abstracts from matching frictions which are left for future research.
Non-bank finance					
Dempsey (2025)	The paper addresses the question of the impact of raising banks' capital requirements on macroeconomic aggregates, banks' and non banks' lending and banks' default.	Two side heterogeneity: ex-ante and ex-post heterogeneous firms, ex-post heterogeneous banks in network.	Value function iteration, combined with aggregate uncertainty.	The ability of the model in succeeding along matching moments and the combination of random matching of firms with banks as well as the presence of short lived ex-ante heterogeneous firms.	The model assumes banks to coexist with non banks in a low market concentration which can be overcome with the introduction of an oligopolistic structure.
Bank runs					
Goldstein et al. (2024)	The paper studies the linkage between bank heterogeneity and financial fragility, showing that a more heterogeneous banking sector enhances stability in the system.	Ex-post heterogeneous banks in bank assets.	No computational solution provided, the model is static.	Exemplifying coordination problems through within-bank strategic complementarity and cross-bank strategic complementarity.	The model can incorporate bank runs through a dynamic incomplete market economy shedding lights about runs at the individual bank level.

NOTE. See associated note at Table 2.

Article/Paper	Purpose	Type of heterogeneity	Solution Method	Strengths	Limitations
Heterogeneous return risk					
Jamilov (2020)	The paper analyses the trade-off between optimal regulation, systemic risk and financial stability to quantitatively assess policy responses to different monetary policies.	Ex-post heterogeneous banks in the returns on capital.	Value function iteration for the banking sector, time iteration for the household sector.	The model can nest the representative bank model in a special case.	Ad-hoc introduction of the nonlinearity in the network evolution.
Jamilov and Monacelli (2025)	The authors studies banking crises driven by an aggregate shock to the capital accumulation equation and the cyclicalilty of the main banks' variables.	Ex-post heterogeneous banks in the returns on capital.	Value function iteration for the banking sector, linear time iteration for the household sector.	The model matches the first two moments of the main banks' variables and reproduces economic recessions driven by the aggregate shock.	The model does not make a distinction for the type of credits which can provide insights on economic recessions driven by non-performing loans.
Bellifemine et al. (2022)	The paper analyses the transmission of monetary policy	Ex-post heterogeneous banks in the returns on capital.	Projection methods for the banking sector, endogenous grid method for household sector.	The aggregate fluctuations are driven by nominal rigidities.	The model can study the impact of unconventional monetary policy in an environment accounting for global banks.

NOTE. See associated note at Table 2.

Article/Paper	Purpose	Type of heterogeneity	Solution Method	Strengths	Limitations
Banks' financial stability					
Mendicino et al. (2024)	The authors analyse simultaneous default of borrowers and banks due to high borrowers' default rate. Optimal bank capital requirement is found to be larger than standard.	Ex-post homogeneous bank/island.	Perturbation solution method.	The ability of the model to reproduce untargted moments. Time variation in the volatility of idiosyncratic shocks that can capture important volatility.	Whilst the idiosyncratic shock is island-specific, results are driven by a representative bank. Essential to capture the model nonlinearities through global solution methods.
Rios-Rull et al. (2020)	The authors study the impact of capital regulation on banks' default. Relaxing the capital requirement leads benefits for small banks reducing their probability of default.	Ex-post heterogeneous banks in credit.	Algorithm not provided.	The crisis period captured by an aggregate bad shock. The ability to smooth the reduction in loan supply of small banks during crisis.	Impossibility to account for entry and exit decision where exit may represent default in the model.
Gödl-Hanisch et al. (2022)	The paper studies the passthrough of monetary policy in the light of the recent rise in bank concentration.	Extrinsic heterogeneity in banks' capital ratio, three-bank model, ex-post homogeneous banks.	The model can be solved with linear solutions.	The findings reveal that a rise in bank market power and bank concentration with capitalisation amplifies monetary policy transmission and leads to a steeper Phillips curve.	Monetary policy and bank concentration are highlighted within a representative banking sector, can be examined in a full heterogeneous banks general equilibrium model.

NOTE. See associated note at Table 2.

Article/Paper	Purpose	Type of heterogeneity	Solution Method	Strengths	Limitations
Coimbra and Rey (2024)	The authors study the concentration of risk in some banks' balance sheet and the implications from endogenous systemic risk and the interaction between monetary policy and financial stability.	Ex-ante heterogeneous banks in risk taking/leverage, value-at-risk constraint.	Policy function iteration.	The model generates bad booms characterised by low cost of funds, worsened financial stability driven by the distribution of leverage right skewed to most risk taking banks.	Impossibility to study the implications arising from aggregate uncertainty. The authors acknowledge the need of a more complex portfolio structure in banks' balance sheet.
		Ex-post heterogeneous banks in loan performance.	Value function iteration adapted to the industry equilibrium.	The model contains a probabilistic structure of size expansion in equity which helps small banks to grow. Stricter capital requirements lead to a decay in loans and increase in concentration.	The model does not account for endogenous loan demand which can extend the model to a general equilibrium.
Goel (2025)	The author studies the optimal bank capital requirement for welfare implications predicting an increase in welfare because of the increase in the mass of banks and capital.	Ex-post heterogeneous banks in quality.	Value function iteration.	Size dependent regulation policies reveal that stricter capital requirements reduce the number of large and less efficient banks, so the concentration of risk.	The model cannot account for actual large banks as without imperfect competition and market power. The model is suited for adding sticky price and their monetary results.

NOTE. See associated note at Table 2.

Article/Paper	Purpose	Type of heterogeneity	Solution Method	Strengths	Limitations
Spatial Heterogeneous Banks					
Corbae and D'Erasmus (2022)	The authors investigate geographic expansion of commercial banks and the diversification that results from a regional aggregate shock brought by the deregulation.	Ex-post heterogeneous banks in deposits.	Value function iteration adapted to the industry equilibrium.	The capacity of the model to deliver the marginal propensity to lend and the spillover effects from the east region to the west that makes the industry more concentrated because of the fall in exit rate of national banks.	The calibration of the model can target parameters to match the marginal propensity to lend in the data. Calibrating these parameters may change the symmetry between parameters and moments.
Oberfield et al. (2024)	The authors explore the spatial expansion of banks due to the 80s-90s deregulation focussing on span on control sorting and mismatch sorting of banks.	Ex-post heterogeneous banks in appeals/productivity over locations.	Algorithm not provided.	The author link the size of banks to the size of cities with mismatch sorting leading large banks to locate in less dense locations with more deposits to reduce their exposure to wholesale funding.	The model can be solved numerically to match the theoretical predictions of the model with the empirical results.
Morelli et al. (2025)	Spatial expansion of banks focussing on funding spreads, deposit markups and risk premia.	Ex-post heterogeneity in demand deposits.	Algorithm not provided.	The authors develop a convoluted heterogeneous banking model which produces a reduction in risk premia and deposit spreads for small counties following deregulation.	The richness of the model abstracts from important considerations to the lending market as well as from a location choice for banks.

NOTE. See associated note at Table 2.

The question that the paper tries to analyse is the implementation of monetary policy through open market operations in order to achieve the target the monetary authority set for the interbank market rate and to what extent frictions in the interbank market impact the real economy through deliberate liquidity injections. In particular, the authors aim to capture the slump of bank lending over the financial crisis 2008-09. Quantitatively, they can match the credit crunch of the US financial crisis, by engendering a worsening of the parameter of the matching efficiency in the interbank market. On policy recommendations, unconventional monetary policies turned out to be necessary and more effective in attenuating the collapse in credit than conventional open market operations.

A valuable contribution of this paper is the modelling of the interbank market with a random search theoretic environment of the Over-the-Counter market in which there are banks in surplus trading funds with banks in deficit. Banks randomly match with a respective counterpart and the probability to match with a bank depends on the aggregate amount of surpluses, S_t^+ , and aggregate deficits, S_t^- . As a result the interbank market tightness is given by $\theta_t \equiv S_t^-/S_t^+$. All the interactions in this market determine the amount of interbank market loans, discount window loans traded with the central bank and the interbank market interest rate. In particular, in the traditional search and matching model of the labor market, the equilibrium wage is a convex combination of what the worker gains by accepting the job and what the worker loses by accepting it.

A similar logic applies in this interbank market. Specifically, the probability for a bank in surplus to match with a bank in deficit, $\Psi_t^+(\theta_t)$, is negatively related to the interbank market tightness, whereas the matching probability for a bank in deficit to match with a bank in surplus, $\Psi_t^-(\theta_t)$, depends positively on the interbank market tightness. As a result of the interactions taking place in this market, the interbank market rate is given by a convex combination between the reserve rate, i_t^m , and the discount window rate, i_t^w , weighted by the endogenous bargaining power of a bank in deficit, $\phi(\theta_t)$, *i.e.*, $\bar{i}_t^f = \phi(\theta_t)i_t^m + (1 - \phi(\theta_t))i_t^w$. By contrast, the allocation for interbank market loans, f_{t+1}^j , and discount window loans, w_{t+1}^j are derived as: $(f_{t+1}^j, w_{t+1}^j) = -s_t^j(\Psi_t^-(\theta_t), 1 - \Psi_t^-(\theta_t))$ if $s_t^j < 0$, and $(f_{t+1}^j, w_{t+1}^j) = -s_t^j(\Psi_t^+(\theta_t), 0)$ if $s_t^j \geq 0$. Essentially, s_t^j represents the surplus of individual banks which can be positive or negative.

The interesting result arising from this market is that all the activities in this market can be encoded in one single object that is the liquidity yield function that represents all the frictions present in this market. Consequently, the liquidity yield function is the main object affecting the aggregate quantities. This liquidity yield function takes the following form: $\chi_t(s_t^j) = \chi_t^+(s_t^j)$ if $s_t^j \geq 0$ and $\chi_t(s_t^j) = \chi_t^-(s_t^j)$ if $s_t^j < 0$. The terms $\chi_t^+(s_t^j) = \Psi_t^+(\bar{i}_t^f - i_t^m)$ and $\chi_t^-(s_t^j) = \Psi_t^-(\bar{i}_t^f - i_t^m) + (1 - \Psi_t^-(\bar{i}_t^f - i_t^m))i_t^w$ represent the slopes of the liquidity yield function and therefore the respective yields for a bank in deficit and a bank in surplus.¹⁴

do not depend on the distribution of banks.

¹⁴The same implementation of the interbank market but in an model with heterogeneous households is provided in [Bigio and Sannikov \(2019\)](#).

A point worth emphasising is that the modelling of this interbank market rests on the assumption that banks have no information about the potential counterpart, and only randomly match with each other. It extended the random search interbank market of Afonso and Lagos (2015) by introducing an infinitesimal trader in charge for a small number of orders so to deliver a unique interbank market rate. It might seem an attractive feature, however a more realistic characterization would be to consider a direct search theoretic environment in which banks actually choose the potential counterpart since within a given day banks have information in terms of the banks that are in surplus and in deficit of reserves.

Currently, research lacks a theoretical environment for a direct search of the interbank market able to match empirical measures such as the interbank market rate, the distribution of banks trading in each session and the amount of intermediated funds with monetary policy interactions. In Gigante (2026b) I take a step in this direction by embedding an assignment matching game into the liquidity market where banks decide to match based on surplus dominance.

The inclusion of the search theoretic interbank market in a general equilibrium context came at the expense of simplifying the other sectors of the economy. The approach of the authors remains compelling. Consider an economy with a unit mass of heterogeneous banks indexed by j . The problem of the banks consists of two stages: the lending stage and the balancing stage. In the *lending* stage, banks enter the economy with some initial level of portfolio illiquid loans, b_t^j , reserves held for precautionary motive, m_t^j , government bonds g_{t+1}^j , and deposits, d_t^j , and they only have to maximise and choose their real dividend, c_t^j , and assets described above, subject to their budget constraint and their capital requirement constraint: $\tilde{d}_{t+1}^j \leq \kappa(\tilde{b}_{t+1}^j + \tilde{m}_{t+1}^j + \tilde{g}_{t+1}^j - \tilde{d}_{t+1}^j)$.

In the *balancing* stage banks enter with the choices coming from the lending stage $\{\tilde{b}_{t+1}^j, \tilde{m}_{t+1}^j, \tilde{g}_{t+1}^j, \tilde{d}_{t+1}^j\}$ and are subject to deposit withdrawal shocks, ω_t^j , affecting the end of period balancing stage of deposits held by banks, $d_{t+1}^j = \tilde{d}_{t+1}^j(1 + \omega_t^j)$. The deposit withdrawal shock, ω_t^j , has mean zero hence it is averaged out across all banks, therefore, all randomness is netted out. Moreover the common distribution to all banks, Φ , is such that $\omega_j \in [\omega_{\min}, \infty)$ with $\omega_{\min} \leq -1$. In this stage banks must respect the reserve requirement given by $m_{t+1}^j \geq \rho d_{t+1}^j$. This stage will give rise to a flow of banks surplus or deficits and bank reserves, whereby banks will then have access to the interbank market described above.

The rest of the economy for firms and households is represented by quasi-linear preferences which, therefore deliver demand schedules and output that are constant elasticity functions. Two points are worth mentioning. First, this is of course, a very strong assumption, since in reality preferences are hardly seen as quasi-linear. However, these preferences make the numeraire good to enter linearly in the utility function and to be valuable which means that compensation to keep the utility constant is possible. Consequently, there is going to be no wealth effect and since utility is an increasing function of the non-numeraire good then the marginal rate of substitution depends only on the non numeraire good. This property allows the authors to make all Euler

equations non-forward looking by fixing one of the marginal utilities to unity.

Second, on the bank side preferences are homothetic and therefore this property allows them to aggregate the model, making equity the only individual state variable without having to keep track of any distributions representing macroeconomic variables which are then simple aggregates since the model then features a representative bank.

The main result of the paper rests on having linked monetary policy through the following banks liquidity premia:

$$\begin{aligned} R_t^b - R_t^m &= \bar{\chi}_t^+ + (\bar{\chi}_t^- - \bar{\chi}_t^+) F(\omega^*) \frac{\mathbb{E}_\omega[(R_t^e)^{-\gamma} \mid \omega < \omega^*]}{\mathbb{E}_\omega[(R_t^e)^{-\gamma}]}, \\ R_t^g - R_t^m &= \bar{\chi}_t^+, \\ R_t^d - R_t^m &= (1 + \rho)(R_t^b - R_t^m) + \left(\frac{R_t^d}{R_t^m} - \rho \right) \frac{\text{COV}_\omega[(R_t^e)^{-\gamma} \bar{\chi}_t, \omega]}{\mathbb{E}_\omega[(R_t^e)^{-\gamma}]} - \mu, \end{aligned} \tag{1}$$

where, R_t^e , represents the return on equity, and μ , is the Lagrangian multiplier on the capital requirement constraint, whereas R_t^b, R_t^g, R_t^m and R_t^d are the real counterparts of the interest rate on loans, government bonds, reserves and deposits, respectively.

The authors study two limiting cases: the Walrasian interbank market and the frictional interbank market. The Walrasian interbank market, that holds when banks are in deficit and have full bargaining power, $\phi(\theta_t) = 1$, or when banks are in surplus and have full bargaining power, $\phi(\theta_t) = 0$. In the latter case, there is an excess of reserves deficits or scarcity of liquidity and since the market is tighter, then the interbank market rate equals the discount window rate.

In the former case, there is excess of liquidity in the system, the probability for a bank in deficit to meet with a potential bank surplus is high and the bargaining power of a bank in deficit is close to one, therefore the interbank market rate equals the floor of the system that is the interest rate on reserves. In this case liquidity in the system cannot be adjusted and the interest rate can be traded either at the floor or at the ceiling of the system and it is indeterminate for zero value of liquidity. By contrast, the frictional interbank market delivers a smooth liquidity yield function and allows the central bank to set any target for the interbank market rate and achieve it with any level of liquidity or by changing the corridor rates. Any of the two strategies will have real macroeconomic effects through the liquidity premia in equation (1).

The first equality in equation (1) is the liquidity premium on loans which is risk adjusted by the interbank market return represented by the liquidity yield of a bank in surplus, $\bar{\chi}_t^+$, and the liquidity yield of a deficit bank, $\bar{\chi}_t^-$, weighted by the risk aversion correction. Banks are said to be satiated when this premium is zero. The monetary authority by changing the nominal rates on reserves and discounted window loans can affect the liquidity slopes of the liquidity yield function and in turn the return on loans banks can set. In particular, when the interest rate on reserves is low, deposits are costly because the cost of insuring against any withdrawal shock increases.

Therefore, the capital requirement constraint is not binding and any increase in the interest rate on reserves will stimulate lending because in this situation loans are complement to reserves. This occurs since as soon as the interest rate on reserves increases then the incentive for banks to hold liquidity increases, however since the incentive to issue deposits increases as well, the greater the incentive to issue deposits, the greater the stimulus to lending through the increase in the interest rate on reserves. On the other hand, only in the particular case in which the capital requirement constraint binds then the interest rate on reserves is high. This implies a lower liquidity premium on loans since the increase in the interest rate on reserves increases liquidity and it is not sufficient to compensate the increase in the interest rate on loans. It leaves reserves and loans to be substitute goods which means any increase in the interest rate on reserves will reduce lending.

This fact implies the behavior of the interest rate on reserves is non monotonic and depends on whether capital requirements are binding or not. The second equality in equation (1) is the liquidity premium on government bonds that yields the same premium on reserves for a bank in surplus. The last equation is the liquidity premium on deposits that is an increasing function of the liquidity risk premium as the amount of deposits increases. Its presence gives banks endogenous risk aversion due to the concavity of the liquidity yield function and because of it then the liquidity premium on deposits can be positive or negative.

Further points are worth to be addressed. The model is a model of heterogeneous banks, but it delivers an ex-post homogeneous bank. Indeed, the authors skillfully show that the bank's problem features aggregation by relying on the theorems of homogeneous functions by [Alvarez and Stokey \(1998\)](#), and despite the non-linearity caused by the non-linear liquidity yield function the bank's budget constraint is shown to be linear in equity, the banks' policy functions are linear and, therefore, the overall model can be solved *as if* facing a representative homogeneous bank.

As a consequence of this aggregation result, the assumption of no persistence on the idiosyncratic shock on deposits makes all states to be recurrent and therefore is not possible to draw the line between individual banks initial conditions and the deposit withdrawal shocks which both can have significant policy implications. Nonetheless, for the way the interbank market was treated and introduced into a general equilibrium context, for having endowed banks with concave preferences over their dividend payments and, for the positive analysis of the liquidity management problem of the banks without losing the essence of the other sectors of the economy, the paper remains a valuable contribution to the literature of the banking system and monetary policy actions.

5.2 Heterogeneous Banks, Money and Collateral

Studying the liquidity management problem of the banks with monetary policy interventions is also [De Fiore et al. \(2026\)](#) who explain the envisaged decline in uncollateralized funds and increase in the threshold of accepted collateral through frictions in the bank liquidity management. They

focus on the interplay of multiple binding constraints on banks. In particular, the policy relevant constraints are the liquidity constraint on deposits as in [Bianchi and Bigio \(2022\)](#) in equation (??), that banks can counteract by trading funds in the interbank market, and the leverage constraint of the type studied in [Gertler and Kiyotaki \(2010\)](#) and [Gertler and Karadi \(2011\)](#).

The main technical insight in [De Fiore et al. \(2026\)](#) on the interactions of the five inequality constraints (a lending constraint, a collateral constraint and three short sale constraints) without imposing any of them to be slack and/or binding is crucial for macroeconomic policies and depends of the specification of the parameter space. From a positive analysis the presence of these constraints explained the decline in the access to the interbank market funds (unsecured or uncollateralized) and a contemporaneous increase in the access of collateralized market (government bonds and money). This in turn raised their two collateral haircuts, the constraint imposed by the central bank and the one coming from the private sector, with the former remaining largely below the latter.

The steady-state analysis in [De Fiore et al. \(2026\)](#) that results from this economic environment by either the number of banks matching in the interbank market or by varying the private sector haircut on collateralized bonds points towards banks deleveraging on deposits. Banks reduce them in order to lower liquidity risk, and move towards increase the holding of secured liquid assets such as bonds and money, in conjunction with underscoring the fall in banks' loan supply. In these circumstances, a tight liquidity constraints gives a fall in loans and output because the leverage constraint turns out to be slack. The only way to prevent a moderate decline in lending, and therefore to relieve the liquidity constraint, is by expanding the central bank balance sheet through injections of liquidity in the economy which would impede a slack leverage constraint.

In my view since the economy is populated by bankers making profits, then one can see the economy with bankers to be isomorphic to a model with a continuum of banks acting in a competitive environment.

In order to create frictions in the interbank market and to give a role for the existence of the interbank market, banks had to be subject to two stages, so-called morning and afternoon stages. The main characterization for the commercial banks is that they can be of two types, connected and unconnected. I refer to the paper for the technical details of the model. Here, it suffices to say that in the *morning* stage, banks discover their type. If they are connected with probability, ξ_t , they can have access in the afternoon to the market of unsecured loans by borrowing. If with probability, $(1 - \xi_t)$, banks are unconnected then in the afternoon they can obtain funds only by exchanging collateralized government bonds. In the *afternoon* stage banks experience a liquidity shock in the form of deposit withdrawal shocks, and the presence of unconnected banks unable to borrow loans during this stage to cover the liquidity needs call, for a constraint called the afternoon constraint by which these banks must hold bonds in order to counteract the liquidity risk.

Despite the presence of these occasionally binding constraints, all of them are linear and in addition to that, what makes the problem tractable is what the authors call the 'big family' assumption of banks by which at the end of the afternoon stage, these banks meet, they equate their positions and begin the next period with the same amount of assets and liabilities. One can think of a two-bank model since all assets, liabilities and value functions are aggregated by weighting their values according to the probability of banks of being connected and unconnected, while at the same time keeping caution of the fact that these probabilities are changing over time so the weight in the population of banks varies. As a result of this big family assumption the distribution of banks is degenerate, thereby allowing for the analytical characterization of the steady state examination which would not have been possible otherwise, if the aggregate objects were depending on the bank distribution.

The model is already rich in many respect, however, it is natural to ask whether one should expect similar policy implications to hold when allowing at least one constraint to be non linear. For example, convex adjustment costs in capitals, by introducing non linearity in the leverage constraint, may render the constraint less likely to bind when liquidity provisions are injected by the central bank. Therefore, the moderate decline in loans and output found in the paper under this policy may be less likely to occur.

Both models by [Bianchi and Bigio \(2022\)](#) and [De Fiore et al. \(2026\)](#) focus on the bank lending channel of monetary policy with the role to cast the fall in loan supply experienced after the financial crisis. They point towards a breaking of the bank lending channel of monetary policy found in the data, but as outlined above neither of both models give rise to an ex-post distribution of banks. Since heterogeneity matters for macro aggregates then both models lack a representation of the macroeconomy through distributions and therefore cannot talk about unequally distributed liquidity across banks, which bank was more affected following a reduction in the unsecured market transaction as in [De Fiore et al. \(2026\)](#) or benefited the most following the policy actions of the Federal Reserve post 2008 as in [Bianchi and Bigio \(2022\)](#), as well as welfare analysis of gainers and losers banks. Some ways to answer these questions can be by emphasizing the relevance of individual bank balance sheets characterised by non linear adjustment costs as mentioned above, and the individual bank's attribute for deposits to depend endogenously on the idiosyncratic shock.

5.3 Bank Heterogeneity and Bank Capital Requirements

In this section I describe models that focus on banks' credit risk differently from the models described above. These models deliver an ex-post distribution of banks that related to the credit risk allows for a different characterization of the economic environment.

The path-breaking work by [Corbae and D'Erasmo \(2024, 2021\)](#) offer another approach to study heterogeneous banks within a banking industry dynamics and contains several features, that I touch upon below, which make the model compelling. The question that they try to

answer is the impact of the introduction of Dodd-Frank Wall Street Reform and Consumer Protection Act that imposed regulatory capital and liquidity requirements to banks, on their lending rate, exit rates and market shares. They analyse banks' concentration by engendering a change in the banks' capital requirements imposed by Basel III regulation and find that an increase in banks' capital requirements from 4% to 6% leads to more concentration in the market through an increase in exit rates, in addition to the fall in loans' supply and a consequent rise in the lending rate.

The paper develops a model of banks' entry and exit dynamics with banks' market power. The strength of the assumption of entries and exits in the market allows the authors to obtain an endogenous size distribution of banks affecting all the equilibrium objects. Normally, in a model of heterogeneous agents, without entries and exits, the model still delivers a distributions of wealth affecting the equilibrium prices and quantities, but this distribution is generated exogenously by virtue of the idiosyncratic shocks affecting agents differently.

The economic environment features imperfect competition amongst banks. This imperfect competition à la Stackelberg, consists of one big bank who is the leader in the loan market and acts consistently with a Cournot equilibrium, having more information than small banks who are the followers in this market. Consider the economy beginning in period t in which after the aggregate productivity shock, z_t , takes place banks' assets and liabilities are realized. Bank i receives profits, $\pi_{i,t}^\theta$, loans, $l_{i,t}^\theta$, liquid assets, $A_{i,t}^\theta$, deposits, $d_{i,t}^\theta$ and networth, $n_{i,t}^\theta$, where θ identifies the bank's specific type. Once the aggregate shock has been realised, the bank can choose to stay or to leave the market. If it decides to stay, the bank matches with some depositors randomly and then make decision over future deposits subject to a capital requirement constraint, δ_t^θ , that follows an exogenous Markov process, $G^\theta(\delta_{t+1}^\theta, \delta_t^\theta)$. Banks can therefore hold different capital buffer because of the volatility in their funding.

The incumbent bank is then subject to three occasionally binding constraints. There is a capital requirement constraint with a bank specific quadratic cost function depending on the amount of loans $k_i^\theta + c_i^\theta(l_{i,t+1}^\theta)$, a risk weighted capital requirement, $\mathbb{E}_t[n_{i,t+1}^\theta] \geq \varphi_t^\theta(\omega_{i,t}^\theta l_{i,t+1}^\theta + \omega_{A,t}^\theta(A_{i,t+1}^\theta + \mathbb{E}_t[\pi_{i,t+1}^\theta]))$, which contains weights φ_t^θ for the capital requirement and on loans and liquid assets that are given by $\omega_{i,t}^\theta$ and $\omega_{A,t}^\theta$. The presence of these liquid assets can then justify why banks would like to hold equity above the minimum requirement imposed by the regulation, since the presence of idiosyncratic shock gives banks the scope for having precautionary behavior. Moreover, the assumption of imperfect competition and entry in the market both imply that profits are not going to be zero in equilibrium.

Lastly, the liquidity constraint is represented as: $\varrho_t^\theta d_{i,t+1}^\theta \leq A_{i,t+1}^\theta + \pi_{i,t+1}^\theta(z_C)$, where, ϱ_t^θ , is the state dependent liquidity constraint. The incumbent bank will then maximise its net dividend payment by taking into account that their decision over loans supply will influence the interest rate on loans taken as a given by small banks. There is an agency problem between banks and firms that is embedded as banks being myopic in their discounting over net dividends

through the parameter, γ , implying that the new modified incomplete market condition is now given by $\gamma\beta(1 + r_t^A) < 1$, where r_t^A is the return on liquid assets.

While I focus on a competitive environment, in Gigante (2026a, 2026b) I postulate a liquidity regulation that takes explicit account of the banks' inflow/outflow of deposits which changes the distribution of banks' liquidity. In Gigante (2026b), this liquidity regulation is submodular in deposits and is interacted with a leverage constraint that works as a loan to deposit ratio constraint. The joint interaction of these constraints modifies the equilibrium distribution of the market.

The entrant bank will decide to enter only if the value from entering net of the entry costs is weakly greater than zero, namely $V_{e,t}^\theta(z_t, \mu_t^\theta, z_{t+1}, M_{e,t+1}^\theta) \geq 0$, where $M_{e,t+1}^\theta$ is the mass of entrants and μ_t^θ characterizes the cross-sectional bank size distribution, that is influenced by the number of new entrants in the market. I refer the reader to the paper for the characterization of the rest of the economy consisting of homogeneous risk neutral households and risk neutral entrepreneurs subject to an agency problem.

The insight coming from the model is that there is a countercyclical amplification mechanism by which the larger the concentration in the industry the larger the downturn. This is due to the fact of the endogenous size distribution that makes entry gradual, namely when a recession hits the economy a large quantity of small banks exit but this exit is not compensated for by the same amount of small banks entering the market. Moreover, the higher the capital requirements the larger the short run welfare gains compared to the long run. Additionally, the paper convincingly matches quantitatively the data. Nonetheless, monetary policy is introduced in a very ad-hoc way by assuming that the interest rate central banks have control to is the interest rate on external debt finance, but without an actual monetary authority nor a welfare maximising policy analysis; that both constitute valuable avenues for future research.

As clearly pointed out from the authors, banks in reality merge and increase their size. A possible way to account for this is to include a jump process in the AR(1) specification of the liquidity constraint. Then, an explicit formalization for the monetary authority existence in addition to this enriched idiosyncratic process can provide a concrete credit channel that interacted with the idiosyncratic process for mergers can provide further observation in terms of the size and magnitude of change of the external finance premium when the monetary authority implements any open market change in the interest rate. Furthermore, big banks in the model are captured by one single bank that is the leader in the market. The model cannot account for a small mass of big banks which seems more an attractive feature as it allows to characterise the few big banks evidenced in the US banking system. In this way, by dropping the arbitrage condition $r^D = r^s$, the small mass of big banks can endogenously give rise to systemic default of SIFIs without occurring through the riskiness of entrepreneurs' projects.

In Corbae and D'Erasmus (2020) the authors keep a close environment to Corbae and D'Erasmus (2021), but they differ by considering three types of banks heterogeneous in their

size: national, regional and local banks. The larger the banks the larger the mean of their funding base and the lower their volatility because of risk diversification. They build an environment with loan market power in a Cournot oligopolistic competition to study the impact of the regulatory policy of Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994 that is interpreted as a reduction in the cost of expanding the funding base.

Similarly to [Corbae and D'Erasmus \(2021\)](#) the endogenous size distribution of banks is brought by the idiosyncratic shocks to deposits and entry and exit dynamics. However, in contrast to [Corbae and D'Erasmus \(2021\)](#) the authors neglect aggregate uncertainty and the Stackelberg assumption in the banking sector. The choice is made to focus on the deposit capacity accumulation through the transition probability matrix that incorporates the possibility for banks to climb a deposit ladder and expand in size. In particular, a small bank may become over time a regional bank, and a medium regional bank may become a national bank. The increase in the deposit ladder, is driven by the investment technology in ι that is bank type specific and time dependent. This investment is interpreted by the authors as merger and acquisition that although occurring within the investment of some units of ι , the process is still exogenous and does not involve an endogenous mechanism. Likewise, the model neglect transitional dynamics pre and post the reform of the regulatory framework and all dynamics is envisaged along movements between two steady states.

The key predictions of the model are the following. First, the distribution of small banks is decreasing in size and second, pre-reform only small banks exit the economy, as regional banks are too costly to acquire by expansion of national banks. The model can account for salient moments of the data such as the deposit and loan market share of the top 4 banks.

Since the data suggest a rightward distribution of banks' size, [Corbae and D'Erasmus \(2022\)](#) highlight the need of a power law distribution to better account for the distribution of banks' size. Likewise, [Bremus et al. \(2018\)](#) emphasise the relevance of a power law distribution and develop a partial equilibrium model of banks and firms to study the granularity of bank size heterogeneity. The granularity concept, developed in an influential work by [Gabaix \(2011\)](#), highlights that idiosyncratic shocks to banks' cost of intermediation do not average in the aggregate but have an impact on macroeconomic outcomes such as credit and output growth through the interest rate on loans. They propose a discrete choice model with market power in the loan market where large banks compete in a Bertrand-style over the interest rate on loans. Thus, they propose an index of concentration and granularity on banks' intermediation as the variable cost of lending.

The endogenous loan markup varies with the idiosyncratic intermediation cost. An endogenous fat-tailed power law distribution of bank size is a necessary condition to obtain enough dispersion in it. To this aim the authors truncate the distribution from above and from below to preserve a non degenerate distribution and to keep the interest rate on loans above the interest rate on deposits. Whilst this environment is a step forward the characterisation of banks' size distribution and because the model aims to capture large and small banks, it might be reason-

able to adapt the granularity concept with the power law distribution to an environment that contains imperfect competition with a Stackelberg bank and followers banks when banks' size is the focus of investigation.

[Dempsey \(2025\)](#) develops a heterogeneous banks model with non-bank finance to study the impact of heightened bank capital requirements and the simultaneous effect of perfect competition of the non-bank sector on banks' loans since higher banks' capital requirements may reduce lending while decreasing risk in the banking sector. The model contains two-dimensional heterogeneity between firms and banks and firm type volatility in productivity as well as aggregate productivity shock.

By combining two moral hazard specifications, managerial diversion and banks' risk shifting, the model can influence not only the quantity of investment but also its riskiness, thereby delivering positive investment also in the event in which the manager does not keep their promises. The paper focusses on the transmission mechanism that arises from the loans in the non-bank sector, *i.e.*, banks extending loans to non-corporate firms, and subsequent impact on banks' capital requirements and macroeconomic aggregates. Higher capital requirements lead banks to build higher capital buffers made of retained earnings and less dividends which in turn reduces the number of constrained banks in equilibrium. However while bank lending decays, non-bank lending rises due to a substitutability effect for corporate firms, whereas non corporate firms are subject to credit rationing. Therefore, financing decisions concentrate in the non-bank sector while investment remains centered in the corporate firm sector.

Endogenous entry and exit dynamics for banks heterogeneous in the level of networth matches them with a random fraction of firms, where banks choose the loan price and monitoring intensity. Monitoring operates to address firms' moral hazard with the manager who may divert part of the profits from the shareholders and creditors for themselves. Profits are allowed to be negative in which case a firm default and creditors recover a portion of the undiverted profits. Banks finance operations with networth, by issuing equity and risk free insured deposits which carry an insurance fee and they are subject to a capital requirement such that equity needs to be larger than a share of risk-weighted assets.

Banks obtain finance by a representative household who invests insured deposits and invest in firms' corporate bonds. A government levies tax and balances the budget of deposit shortfalls of failed banks. In choosing the lender type, firms maximise their value which is affected by a type I extreme value preference idiosyncratic shock. Since firms are subject to default, in the non-default state lenders who map into the firm with their loan demand receive full repayment, whereas in the default state lenders obtain partial repayment with the output non diverted.

Heterogeneous banks choose exit at the beginning of the period before knowing the type of the firm they are randomly matching with. They maximise the expected discounted value subject to the resource constraint which contains the deposits net of insurance fees financing loans and dividends. Importantly, as firms solve a static problem and they are short lived, then

their future distribution does not depend on the current value of the distribution but instead on banks' future distribution which facilitates the computational solution.

The result rests on the timing of the idiosyncratic shock as well as on the nature of the idiosyncratic preference shock led by an extreme value distribution, which allows ex-ante heterogeneity in firms' type to survive also ex-post. Indeed, by reducing the scale parameter the idiosyncratic preference shock increases firms' elasticity of loan demand thereby raising competition between banks and non-banks. A representative household who own banks, through the stochastic discount factor, makes choices of consumption, labor, loans, non bank loans and bank equity. Lastly, a government provides deposit insurance when a bank cannot meet its obligations upon exiting which is financed by taxing labor and profits and charging deposit insurance fees to the household.

By estimating model moments to match empirical counterpart moments through simulated method of moments, the model succeeds in several dimensions such as firms and banks' leverage, share of firms' debt and banks' failure rate. The impact of a small rise in banks' capital requirement leads to a substantial increase in networth which is obtained by raising equity and reducing deposits since the capital requirement almost always bind. By retaining more earnings, capital buffers rise and modestly reduce the default rate. Therefore, raising bank' capital requirements have a small effect on aggregate variables unless changes occur on the modelling structure which embeds a less competitive environment in the banking sector.

The model constitutes one of the most promising models dealing with non-bank finance in a context of heterogeneous banks and firms. The model can be further improved by embedding market frictions that account for more banks' concentration in the market for loans such as an oligopolistic banking environment or monopolistic competition which may provide the ground for larger quantitative responses of aggregate variables such as output, leverage, welfare and banks' default upon raising banks' capital requirements.

5.4 Bank Deep Habits

A similar framework that delivers an endogenous size distribution of banks in a dynamic banking environment is given in [Dempsey and Faria-e Castro \(2025\)](#) who focus on the market power of banks in having long-lasting relations with their lenders. The question they try to address through a quantitative model of bank industry dynamics is the trade-off that banks face in their pricing decisions. Higher interest rates on loans bring greater profitability in the short run, but in the long run they have the opposite effect, because high interest rates on loans reduce the future number of borrowers, thereby reducing banks' customer capital and decreasing profits since the borrower lender relationship is typically long-lasting.

These long-lasting relationships are conceived by [Dempsey and Faria-e Castro \(2025\)](#) as a form of bank market power and they are modelled as deep habits in the loan demand of

the borrower.¹⁵ In this model not only is the loan habits relationships drawn from a Markov process exogenously, but also the loan return that the bank will receive. In Gigante (2026b) I introduce convex adjustment costs in illiquid assets without any habits relationship to focus on the redistribution of banks' assets.

The amount of customer capital that banks own determines the speed of recovery after a negative shock to networth. Banks owning more customer capital experience a lower drop in their market value and recover more quickly compared to banks with low customer capital. This occurs because banks with more customer capital are able to keep charging a higher loan rate relatively to banks with less capital, therefore this point towards the presence of cross sectional variations amongst banks following a shock that a representative bank model would hide and will not be able to capture.

Quantitatively, they find that after a negative shock to bank networth the higher the persistence in the habits formation on loans, the larger the fall in customer capital in the short run but over time, when aggregate customer capital reaches its largest negative value, average networth increases to its largest positive value. This intuitively means that banks are not necessarily punished for the rise in the loan interest rate and fall in profits driven by the fall in customer capital. Moreover, the supply of loans and interest rate on loans are higher for banks with high customer capital compared to banks with small customer capital. These banks are also the ones that recover more quickly since their effect on networth is smaller than the decline that small banks experience.

The model is tested to assess the fall in the Lerner index (bank market power) and the fair value of loan portfolio, both used as proxy measures for the US over the period 2004 – 2009 whose distributions are both skewed to the right representing banks in the 95th percentile. Using cross-sectional data on US bank holding companies they find a statistically significant positive relationship between changes in market loan and changes in customer capital, implying that the loss in bank's loan market value over the financial crisis was accompanied by a large loss in bank's customer capital. By contrast, they found a statistical and significant negative relationship between changes in post crisis market value and changes in customer capital over the crisis period. The stronger capitalisation was led by banks that used mostly customer capital.

The model is partial equilibrium and a general equilibrium effect is present through the price of loans and its spillover effects. Therefore, the model cannot draw predictions in terms of consequences to the overall economy. Since the focus is on the propagation of the shock, it is important to endogenise exit choice to capture loan dynamics with the mass of entrants changing over time.

¹⁵The deep habits formulation was originally proposed by Ravn et al. (2006) and Ravn et al. (2010) to evaluate the business cycles and the effects of monetary policy shocks.

5.5 Bank Heterogeneity and Liquidity Regulations

In this Section I consider an economic environment that I developed in Gigante (2026a). This model of heterogeneous banks allows to study the effect of liquidity regulations on banks' size expansion, sorting of banks by type, the accumulation of banks' liquidity due to ex-ante liquidity regulations, as well as the design and implementation of liquidity regulations on liquidity crises. The model is amenable to different characterisations of the analysis of bank heterogeneity in depth. The distinctive features of the model for the banking sector are *i*) the idiosyncratic nature of deposits and *ii*) a liquidity regulation designed as a function of the level change of deposits and *iii*) a liquidity requirement that gives incentives to banks to insure by building a buffer of liquidity.

The model is as follows. Assume an infinite time closed economy model with $t = 0, 1, 2, \dots$. The economy is inhabited by a continuum of heterogeneous banks living in a competitive banking sector. Banks have preferences over a stochastic sequence of dividends income, $c_{i,t}$, given by:

$$\mathbb{E} \sum_{t=1}^{\infty} \beta^t u(c_{i,t}), \quad (2)$$

where $\beta \in [0, 1]$ is the banks' discount factor. Banks invest in liquid assets, $\ell_{i,t+1}$, which yields a gross return r_ℓ , they invest in illiquid assets s with gross return r_s and their funding is composed by deposits, $d_{i,t}$ that pay the gross interest rate r_d . As occurs in reality, it holds the following condition $r_\ell < r_d < r_s$. Deposits are subject to idiosyncratic shocks and they are driven by a Markov process, $\lambda(d_{i,t+1}, d_{i,t})$. Consequently, banks cannot trade assets in any given period to insure against the withdrawal risk on deposits.

Banks are subject to an ex-ante intraperiod liquidity requirement that they internalise in their programme. The beginning of period liquidity constraint is given by:

$$\ell_{i,t} + (d_{i,t+1} - d_{i,t}) \geq \phi, \quad (3)$$

where ϕ is an exogenous liquidity requirement. Liquidity regulations generate an expansion in banks' size. Consequently, banks are subject to the liquidity regulation prior to the realisation of the deposit outflow, in addition to letting liquidity regulation in the level of deposits thresholds instead of a liquidity ratio. Banks cannot raise external funding but they can store equity through accumulated retained earnings. Therefore, the beginning of period banks' equity is given by:

$$a_{i,t} = \ell_{i,t} + s + \mathcal{P}_{i,t} - d_{i,t}, \quad (4)$$

where $\mathcal{P}_{i,t}$ are the beginning of period profits. A binding liquidity constraint forces small and medium banks to raise their accumulation towards liquid assets reducing their exposure to the idiosyncratic deposit withdrawal shock. Since the economy is subject to incomplete markets,

the equilibrium outcome driving results is the condition $\beta r_\ell < 1$. The condition guarantees the existence of a well defined distribution of liquid assets and therefore that in a non binding liquidity constraint case banks do not give rise to an ever increasing amount of liquid assets.

Since banks are subject to ex-ante liquidity regulation, the ex-post banks' budget constraint is given by:

$$c_{i,t} + s + \ell_{i,t+1} - d_{i,t+1} = r_s s + r_\ell \ell_{i,t} - r_d d_{i,t} + (r_\ell - r_d)(d_{i,t+1} - d_{i,t}). \quad (5)$$

The budget constraint captures the inflow/outflow effect of deposits through the term $(r_\ell - r_d)(d_{i,t+1} - d_{i,t})$ and makes clear that the withdrawal of deposits is costly for outflow banks since they have to pay the interest rate wedge on the deposit rolled over to inflow banks. The resulting ex-ante liquidity regulation that banks internalise in their programme is:

$$\ell_{i,t+1} + (d_{i,t+1} - d_{i,t}) \geq \phi, \quad (6)$$

and the nonnegativity of dividends consumption requires banks respecting $c_{i,t} \geq 0$. The banks' programme is to maximise equation (2) subject to (5), (6), $c_{i,t} \geq 0$ and the end of period equity $a_{i,t+1} = \ell_{i,t+1} + s + \mathcal{P}_{i,t+1} - d_{i,t+1}$.

There are five points worth taking. First, the inflow of deposits by expanding the deposits set of one bank, it raises the liquidity requirement and therefore it expands liquidity and the liquid assets set. The result is because of the liquidity requirement in the level of deposits instead of being a ratio and, the ex-ante liquidity regulation that imposes them to incorporate the regulation into their beginning of period resources. The value of banks is then increasing in the size of the inflow/outflow of deposits.

Second, the model gives rise to negative sorting of banks whereby small banks with large deposits hold less liquidity than large banks with less deposits. The expansion of banks' size is such that small banks with large deposits can also expand in size, whereas for large banks it holds that the least largest banks can expand in size posing an endogenous bound on the size of banks. Furthermore, depending on banks' initial type, if the realisation of the shock is at the lower bound then inflow banks expand in size, whereas if the realisation of the shock is at the upper bound then banks' size expansion does not occur. Therefore, the extended sets of liquidity and deposits need to be measurable and I show that the banking programme under the expanded set remains measurable.

Third, the next result of the model is that the tighter the liquidity requirement the more equally distributed the distribution of banks' liquid assets becomes. This is a consequence of the stronger incentive for medium and small banks to hold liquid assets than large banks, thereby reallocating liquidity towards medium and small banks. This reallocation of liquidity implies that the fraction of small banks in the bottom percentile of the distribution increases more than the upper tail, driving down inequality.

By imposing ex-post regulation only, the model nests Holmström and Tirole (1998) as a special case in addition to leading to the same theoretical and quantitative predictions in terms of banks' liquid assets inequality of the baseline economy in which ex-ante regulations are imposed jointly with a tightening of the liquidity requirement. Indeed, I show the central equivalence result of the paper that makes the two economies equivalent. Fourth, in the economy with the absence of liquidity regulations, banks' incentives towards liquidity and dividends are not affected by the liquidity requirement. This implies that there is not a change in the total cost on short term debt thereby the distortion in banks' balance sheet resources does not arise anymore. Consequently, the findings pertaining to banks' size expansion and the sorting of banks by type does not hold anymore since without ex-ante regulations banks may fall below the liquidity requirement.

Fifth, pertaining to liquidity crisis the withdrawal shock on deposits delivers contractionary effects on banks' balance sheet variables and dividends. However, the prediction of the model is that ex-ante liquidity regulations reduce the contraction in banks' liquidity and profits by approximately 23% less than the ex-post regulated economy fostering the analysis that ex-ante liquidity regulation help to mitigate the effects of the crisis. Therefore, market incompleteness and ex-ante liquidity regulations set prior to the realisation of the deposit outflow make the banking system more resilient, thereby reducing the probability and episodes of liquidity crises. The findings remain consistent when considering a comparison with the ex-post regulated economy and the economy without regulations.

Whilst the need to analyse liquidity regulations in a heterogeneous banking model made the setting general enough to encapsulate the models outlined in Section 5.1 and 5.3, it can be examined by considering relevant features of the banking sector that deserve further consideration. As pointed out by Gigante (2026a) while the model contains a high return illiquid asset and illiquid assets, they are left exogenous to the programme thereby making impossible to explain the marginal propensity to lend and to confront it with the data. Gigante (2026b) makes a step toward this direction and endogenises illiquid assets to study the marginal propensity to lend.

In particular, Gigante (2026b) develops a theoretical framework with banks' liquid and illiquid assets, adjustment costs in illiquid assets and idiosyncratic deposits withdrawal shocks. The theory places particularly importance to the interaction between balance sheets constraints and adjustment costs in illiquid assets to generate the endogenous market structure. More precisely, the model delivers liquidity constrained banks and leverage constrained banks with the redistribution of assets occurring along the entire distribution of banks. By endogenising how a given mass of banks is reallocated into their size, adjustment costs in illiquid assets and submodular regulatory constraints makes the matching between large and small banks strictly surplus dominant determining the market structure of banks.

The model generates a highly unequal distribution of the marginal propensity to lend between large and small banks, and finds that medium large banks have a 8.5% elasticity of illiquid

assets to a 1% upward change in the interest rate on loans. The findings reveals negative assortative matching in the banking sector that is pervasive when the strength of submodularity in the liquidity regulation rises. Moreover, the theory allows for a rich transitional dynamics. When examining the Bank Term Funding Program, the results suggest that the program would have generated a 1.95% expansion in illiquid assets through a redistribution of 4.51% of liquid assets, and a 77.85% rise in the marginal propensity to lend among banks.

5.6 Bank Heterogeneity in the Return on Capital

Jamilov (2020) incorporates bank heterogeneity in the banking framework of Gertler and Kiyotaki (2010) and Gertler and Karadi (2011) in order to study bank default risk with economic and financial crisis at the individual bank level. The paper addresses the question of whether bank heterogeneity matters for optimal regulation, systemic risk and financial stability. In particular, it quantitatively assesses policy responses of equity injections to banks, direct lending by which a central bank can lend directly to firms at low marginal costs, liquidity facilities representing monetary authority discount window lending to banks and deposit guarantees schemes that close default risks for banks.

In this model there is a moral hazard problem between banks and households that delivers an endogenous limit on the leverage ratio since banks borrow from households through short term debt and accumulate network. The moral hazard, developed in analogy with Gertler and Kiyotaki (2010) and Gertler and Karadi (2011), is in the deposit market by which the bank may have an incentive to divert no more than a fraction of the total value of capital revenues.

The idiosyncratic risk for banks is encapsulated in the rate of return on capital that is embedded as a convex combination between an idiosyncratic and a fixed component. The idiosyncratic risk return delivers ex-post heterogeneity in returns that transmits to banks network. A negative shock to the rate of return on capital implies that banks can enter into insolvency.

Similarly to Corbae and D'Erasmus (2021), they keep the entry and exit dynamics on the bank side and the imperfect monopolistic competition of banks. In particular, imperfect bank competition is embedded by assuming that banks enter the period with no equity and the competitive firm issues claims to banks in return for a capital bundle. Consequently, there is bank market power over differentiated capital goods that are then aggregated according to the Dixit-Stiglitz aggregator capturing imperfect competition between banks in the same style of Blanchard and Kiyotaki (1987). Therefore, the bank has preferences over financial varieties to deliver a constant elasticity credit demand.

The bank-firm relation is captured in analogy with discrete choice theory by which the borrower can choose both the size of the loan and the bank type from which to borrow, representing credit market power at bank level. The individual network of the bank is subject to a quadratic capital cost function representing expenses other than interest rates, capturing increasing or decreasing returns to scale thereby giving rise to a nonlinear evolution of network

as in Corbae and D’Erasmus (2021).

The credit market competition gives rise to an additional term in the leverage specification that is the ratio of individual capital over aggregate capital that depends on the credit elasticity of substitution that multiplies the actual leverage of capital over networth, $\phi = (\frac{k}{K})^{-\frac{1}{\theta}} \frac{Pk}{n}$. This means that because of the Dixit-Stiglitz aggregator, when the degree of credit elasticity, θ , converges to infinity the model becomes a model of credit market competition where the leverage reduces to $\phi = \frac{Pk}{n}$, thereby converging back to the models of Gertler and Kiyotaki (2010) and Gertler and Karadi (2011).

Bank heterogeneity in the model is delivered by heterogeneous marginal costs that are given by the idiosyncratic return on capital times the quadratic scale component on capital. Moreover, the price rule is now determined at the bank level given by the mark up times the marginal costs. Since small banks face low networth, a high deposit rate, and low idiosyncratic realisations then they have a high probability of default. Consequently, marginal costs are decreasing in bank networth implying a decreasing price rule in bank networth. Since marginal costs are decreasing in bank size, the larger the bank the lower the marginal costs. The household problem is standard expect that the return on deposit is risky because of possible bank default and no deposit insurance.

Quantitatively Jamilov (2020) can match the distribution of bank leverage fairly reasonably and obtain the marginal propensity to lend, calculated in this model as the change in capital given a change in networth, which resulted into being increasing in networth and decreasing in the idiosyncratic return on capital. Moreover, consistent with the data, the model can generate a positive correlation between cross sectional bank leverage and bank size because of decreasing marginal costs in bank size. Thus, large banks can take advantage of their lower marginal costs by extending more loans.

Analysing the credit policy responses, the model generates a distribution of banks’ networth skewed to the right and banks with high networth have a greater marginal propensity to lend whose elasticity differs from a representative bank model by a factor of two. It is the presence of this marginal propensity to lend linked to the bank marginal costs that drives the aggregate outcomes. Moreover, the model also delivers concentrated and dispersed distributions of assets, deposit rates, default risk and relative prices. Equity injection policies in this model are efficient if targeted to large banks, whereas direct lending facilities, deposit guarantees and liquidity facilities schemes have the most significant impact if directed to small banks.

The model contains quite a few levels of frictions, whereby the joint interactions of them may overstate the response of one friction compared to the other. Moreover, by fitting the model in many dimensions one always needs to find a trade-off hence sometimes it is difficult to say the single contribution of each parameter into the model. Additionally the dynamics of networth is exogenously set to be nonlinear.

Jamilov and Monacelli (2025) extend the Jamilov (2020) model in two ways. First, by

allowing for aggregate technology shock to the banks' balance sheet, namely the aggregate shock affects the law of motion for capital which then feeds into the return on capital and second, by adding local decreasing returns to scale on capital where banks do not internalise the impact of private loan margin choices on aggregate demand as in [Blanchard and Kiyotaki \(1987\)](#).

The contribution of [Jamilov and Monacelli \(2025\)](#) is having provided additional evidence of bank level heterogeneity documenting some facts at the individual bank level to be matched in the model. Their quantitative exercises characterise and identify episodes of systemic banking crisis as large economic recessions, large reduction in the number of intermediaries and bank leverage.

Regarding the empirical facts, they document leverage ratio distribution and assets that are rightly skewed, though with a small mass of risky banks. In terms of cyclicity, they find a positive significant statistical correlation between bank size and interest rate margin, and a positive correlation between bank leverage and bank size. Plotting the series for bank assets, equity and leverage the first and second moments are pro-cyclical with the business cycle, whereas the concentration is countercyclical. On the other hand, loan margins and default risk have a countercyclical first and second moment, whereas the third moment is countercyclical for both, with caution for the third moment of default risk since it is subject to sample selection.

The model closely follows [Jamilov \(2020\)](#), therefore I refer the reader to the paper for additional specific modelling choices, here it suffices to say that the additional features are the incorporation of aggregate shock in the capital accumulation equation and decreasing returns to scale in individual capital that affects individual banks' balance sheet.

The simulation of the model delivers a shift of the distribution of bank size to the left after a negative aggregate shock to capital. This means that bank size is now centered around zero because of limited liability driven by bank exits from the market due to bank default, therefore the financial sector becomes more concentrated. The distribution of networth shifts also to the left that means that banks small in their book leverage are less risky. Moreover, by exploring bank crises the authors find that with acyclical idiosyncratic risk the larger the economic recession the more modest the financial crisis that follows, whilst countercyclical idiosyncratic risk leads to a joint occurrence of severe economic recessions and financial fragility.

Although the model features credit market power where the types of credit are grouped via an aggregator, it appears an intrinsic heterogeneity that cannot account for the specific variety of loans, *i.e* nonperforming loans. Introducing a more explicit credit heterogeneity can provide insights on the economic recessions driven by this type of loans.

Along the lines of [Jamilov \(2020\)](#) and [Jamilov and Monacelli \(2025\)](#), the need for the joint interaction of market power on deposits and market power on loans lead [Bellifemine et al. \(2022\)](#) to extend the first two papers with nominal rigidities to study the transmission of monetary policy. Credit market power has amplification effects to a contractionary monetary policy shock occurring because of the impact on large banks due to the incompleteness in interest

rate passthrough. By contrast, deposit market power attenuates responses to a contractionary monetary policy shock as this market power increases. The model can be extended to analyse the impact of unconventional monetary policy on global banks.¹⁶

5.7 Bank Heterogeneity and Bank Runs

The influences and effects of banking distress originated by bank runs has a long root in macroeconomics that goes back to [Diamond and Dybvig \(1983\)](#) who model the function of liquidity creation of a bank in a depositor-creditor framework. In particular, they develop a model in which they give an explicit role to the demand of liquidity by banks and the transformation of illiquid assets into liquid assets through liabilities contracts offered to investors.

The presence of this contract gives rise to multiple equilibria despite the presence of a symmetric information set. The first equilibrium is a good equilibrium in which there is full information and banks share the risk of a bank run characterised as pure strategy Nash equilibrium in which one agent withdraws their money and the second agent, anticipating the action, waits until the next period. The second equilibrium is a bank run equilibrium that leads to panic into the economy causing agents to withdraw their money altogether in the first period since they expect the bank to fail.¹⁷

This approach has been taken as a reference for a deep understanding of systemic financial fragility and how to prevent episodes such as financial crises in general equilibrium contexts, some instances are [Gertler and Kiyotaki \(2010\)](#), [Gertler and Kiyotaki \(2015\)](#), [Gertler et al. \(2016a\)](#), [Gertler et al. \(2016b\)](#), [Gertler et al. \(2020b\)](#), [Gertler et al. \(2020a\)](#) and recently [Amador and Bianchi \(2024\)](#), [Amador and Bianchi \(2025\)](#). The distinctive features of these models are two. First, the formation of beliefs about run probabilities that are tied to aggregate economic fundamentals through an exogenous threshold rule, nesting the global game nature in microeconomics, and second, bank runs occur for the entire financial sector.

In [de Groot \(2021\)](#) the bank run problem is microfounded as bank global game through a coordination problem that arises as a result of a sequence of static global games as in [Morris and Shin \(2004\)](#). Specifically, the coordination problem gives rise to a trade-off between the illiquidity premium that banks bear on their balance sheet and the borrower leverage that amplifies the volatility in aggregate macroeconomic variables, in addition to shocks in credit market liquidity. Moreover, the coordination problem generates multiple equilibria of rollover debt that can engender four scenarios: no fragility in which no wholesale banks experience an illiquidity problem, mild fragility in which the only insolvent wholesale banks are subject to

¹⁶Analysing the impact of monetary policy transmission through market power on deposits and loans is also [Wang et al. \(2022\)](#) who develop a structural partial equilibrium model of household, firm and banks to quantify the effect of bank market power and the passthrough of the federal funds rate to deposit and loan rate. They find that expansionary monetary policy may contract lending through the fall in profits in the deposit market that in turn affects capital in a contractionary way. Whilst the model is accounted to be a banking industry, it does not however consider the possibility for banks to endogenously exit the economy.

¹⁷For a narrative explanation of [Diamond and Dybvig \(1983\)](#) model see [Diamond \(2007\)](#).

run, acute fragility characterised by solvent and insolvent banks experiencing a run, and a ‘no rollover’ scenario in which rollover never occurs for retail banks.¹⁸ The model dynamics shows that the Fed credit market policies undertaken as a consequence of the financial crisis were likely more effective in the form of equity injections than direct lending or liquidity injections and that bank runs tend to increase in periods of illiquidity and recessions.

Differently from the multiple equilibria setup analysed in the above models are the frameworks developed in Allen and Gale (1994), Allen and Gale (2004), Allen and Gale (2009) and Uhlig (2010) who consider fundamental equilibria in which partial bank runs (in contrast to bank runs by all banks) are due to heterogeneous beliefs amongst depositors for whom it is rational to run although nobody else takes this action.

Since these models feature a homogeneous financial sector, I draw the attention to the model developed by Goldstein et al. (2024) who consider heterogeneous banks in a static bank run framework. Heterogeneity is incorporated in individual banks’ assets fundamentals through the presence of idiosyncratic and aggregate shocks.

The central question that the paper seeks to address is whether increased level of heterogeneity in banks can raise the stability of the banking sector since homogeneity amongst them might synchronise the fragility of the system caused by bad shocks. In this regard, they modify Diamond and Dybvig (1983) framework so as to allow for endogenous price liquidation and fire sale externalities arising when banks liquidate their assets in a bank run.

Moreover, they take a global game approach à la Morris and Shin (2001) in which incomplete information about the beliefs and actions of the other players and uncertain economic fundamentals lead banks to observe a signal of the state that varies across them according to some noise. Heterogeneous banks are ex-post categorised into two types following an idiosyncratic shocks to bank’s fundamentals, differently from the aggregate shock that affects all banks in the same way. In particular, strong banks are identified as those banks ex-post positively affected by the idiosyncratic shock to banks’ fundamentals, and weak banks are those negatively affected by the shock.

A coordination game arises because of within bank strategic complementarity and cross bank strategic complementarity. The former gives rise to a standard risk for all banks should creditors withdraw their money, whilst the latter implies that the liquidation cost sustained by banks increases with the number of banks liquidating their long-term assets as a consequence of the payment to creditors who have withdrawn their money. By virtue of the beliefs, a coordination problem that characterises global games, the incentive for creditors to withdraw money rises when they believe that other creditors in the economy are undertaking the same

¹⁸A large amount of uninsured deposits may generate bank runs and instability in the financial system with banks defaulting endogenously (Egan et al., 2017). In a model where banks choose the interest rate on insured and uninsured deposits and face difference in the cost of servicing can lead to multiple equilibria. In contrast to Diamond and Dybvig (1983), multiple equilibria in this environment are driven by not only beliefs but also fundamentals. Findings suggest that low capital requirement and below 18% can destabilise the banking system. The model however does not feature incomplete markets.

action.

The insight coming from the model is that the larger the uncertainty surrounding creditors' beliefs, the larger the heterogeneity amongst banks fundamentals the stronger the stability of all banks because the increase in heterogeneity is brought by making weak banks more fragile, thereby rising the discrepancy in banks' fundamentals between strong and weak banks but delivering a more stable system in absolute terms for all banks.

A limitation of this paper is the lack of an extensive theoretical argument of bank runs in a general equilibrium context of heterogeneous banks in which government policies such as credit interventions are guided by an explicit government sector and the monetary authority can bring an additional stabilization mechanism through a welfare maximising criteria for all agents. The majority of the analysis is carried out according to a two-bank model in a static environment. A more systemic consideration of bank run matters in an incomplete market economy in a dynamic context is a fruitful avenue of research.

Contrastingly, a case in point is a dynamic macroeconomic perspective of self-fulfilling bank runs on individual banks that can increase the number of defaulting banks generating an increase in financial fragility as developed in [Amador and Bianchi \(2024, 2025\)](#). The run in this case occurs because of the positive spread between the return on capital and the cost of borrowing which leads to a reduction in the value of a bank. In this case because of the investors' belief about other investors' belief going to run the bank becomes unable to make profits.

Controlling for heterogeneity may promote reasons for banks to coordinate between them and are likely to produce positive feedback to the macroeconomy. However, there has been no theoretical and quantitative research on ex-post bank heterogeneity and their runs in a dynamic context which is needed if we are to better understand why heterogeneous banks undertake runs in a panic situation.

5.8 Heterogeneous Banks, Financial Stability, Business Cycles and Bank Risk Taking

The relevance of heterogeneous banks in macroeconomic models is also stressed in the macro banking literature analysing banks' default at the intersection with banks' capital requirements.

[Rios-Rull et al. \(2020\)](#) focus on banks' credit risk, similarly to [Corbae and D'Erasmus \(2024, 2021\)](#), however they study the interaction of countercyclical capital buffers with banks' defaults by including uninsured wholesale market-based funding whose price represents a premium on bank specific default.

The research question they seek to address is the joint functioning of market disciplines in bank funding that originates when a bank finances some of its assets with unsecured debt and the regulation imposed by Basel III, that requires banks to satisfy a minimum capital requirement and countercyclical capital buffer. The former requires banks to hold capital above the minimum threshold weighted by risk, while the latter instructs banks to build capital buffer in normal times owning more capital during expansion phases and lower capital in period of

financial turmoils and recessions. The violation of these requirements implies that banks need to reduce their dividends and use the distribution of earnings in order to rebuild capital. This strains their normal activity, unless the violation is too large in which case the bank is put in bankruptcy.

Two forces might be at work when market disciplines are in place. First, they can limit bank risk taking by introducing restrictions on price and quantities for uninsured bank funding therefore regulation in this case is needless. Second, countercyclical capital buffer can increase social welfare jointly with market disciplines in case in which the only market discipline reduces the optimal amount of bank loans. To answer this question the authors build a partial equilibrium model of banks and introduce two types of risky assets: long term risky loans and short term risky wholesale funding.

While they describe this is a model of banking industry, this is not analytically developed. An example is in [Corbae and D’Erasmo \(2021\)](#). Incorporating industry dynamics through entry and exit decisions can provide quantitative guidance on actual banks’ default. Loans are idiosyncratically affected and a bank can default because of inability to pay its liabilities or because of preference over the outside option. Because of incomplete markets, the model features ex-post distribution of bank loans affected by increasing and convex costs of issuing loans. In particular, this issuance cost is increasing in networth and decreasing in loans in such a way to obtain increasing returns to scale in bank size. Moreover, the second source of non-linearity in bank networth is driven by wholesale funding costs being increasing and convex.

By simulating the model over crisis periods, where a crisis is defined as an aggregate bad state with a worsening of the loan failure rate, the model predicts a fall in the bank’s loan supply and an increase in funding sources. In addition, banks at the bottom of the capital ratio distribution benefit from an eventual relaxation of the capital requirement constraint during a crisis because these banks are more likely to be constrained by the capital requirements. Therefore, this policy reduces the decline in loan supply of small banks and their probability of default.

Supporting the view of a decay in aggregate credit induced by the rise in banks’ capital requirement is [Mendicino et al. \(2024\)](#) who also find a fall in the probability of twin defaults. Their finding is due to the increase in the average funding cost which leads to higher borrowing costs. The authors study macroeconomic and financial stability similarly to [Rios-Rull et al. \(2020\)](#) however, in [Mendicino et al. \(2024\)](#) banks are affected by firm-borrower solvency. Therefore, they develop a model of banks’ defaults with an intrinsic form of bank heterogeneity. Specifically, the model is a variant of [Gertler and Kiyotaki \(2010\)](#) and [Gertler and Karadi \(2011\)](#) modified to allow for idiosyncratic banks’ loan returns in which twin defaults are thought as a simultaneous high frequency of defaults in both firms and banks.

The idiosyncratic state is island specific, but not bank specific. Therefore, what counts is the wealth distribution of the overall banking sector and not the distribution of individual bankers

since their value functions and policy functions are linear in networth and no ex-post distribution of banks survive in the aggregate. This in turn implies that the island is characterised by a representative bank. However, although the idiosyncratic shock is log normally distributed, the resulting banks' loan returns are a nonlinear function of the idiosyncratic shock. Therefore, because of these nonlinearities in banks' assets returns at the aggregate level the model is able to match important moments observed in the data such as the unconditional correlation of firms and banks defaults in conjunction with the frequency and the severity of twin default crises.

The model underlines the contemporaneous importance of nonlinearities in the asset return function and time variation in the standard deviation of the idiosyncratic shock to generate a high sensitivity of output to twin defaults. This is necessary because the small fluctuations caused to firm default are not sufficient to engender bank default through the sole presence of aggregate uncertainty in total factor productivity.

Consistently with heterogeneous bank leverage in lending, [Gödl-Hanisch et al. \(2022\)](#) show in a three-bank model that the passthrough of the policy rate to the short term lending rates is strong for banks undercapitalised and operating in markets at high bank concentration. The exact nature of the heterogeneity makes the findings based on a representative banking sector. The larger capitalisation and financial stability that result from a rise in bank concentration are delivered with linear solution method. An in-depth characterisation of the banking sector based on an incomplete market economy in a general equilibrium context may reveal more accurate quantitative performance of the impact of bank concentration on monetary policy passthrough and financial stability. Furthermore, a calibration strategy based on targeting moments conditions can provide accurate estimates on monetary policy passthrough.

In line with heterogeneity in banks' leverage, [Coimbra and Rey \(2024\)](#) explain the forces that drive endogenous systemic risk, balance sheet concentration and the interplay between monetary policy and financial stability. Their model features a Value-at-Risk constraint that delivers ex-ante heterogeneity in bank risk taking causing endogenous financial stability. The Value-at-Risk constraint dictates for banks that the probability of a negative return on capital must be smaller than an exogenous parameter that is different across intermediaries. The model predicts a right-skewed distribution of bank leverage since banks with high default probability holds the majority of assets in the economy which worsen financial stability.¹⁹

This performance is generated starting from a low interest rate since the extensive margin of the model is driven by entry of high risk taking banks that will be operating at the binding constraint. By contrast, when the interest rate is already high, further reductions produce an increase in leverage where entry into the market for risky projects is driven by less risk taking banks. This can be thought as monetary policy having a selection effect and a composition effect, and financial stability is brought by endogenous balance sheet changes by keeping equity constant. The model lacks an active role for the monetary authority, therefore a closer inspection

¹⁹The importance of aggregate risk taking when banks are heterogeneous in an incomplete market economy is studied in [Korinek and Nowak \(2017\)](#).

to a nominal model rather than a real one could play a rather larger role for the endogenous financial stability by affecting the cost of funds of financial intermediaries.²⁰ Aggregate uncertainty due to an aggregate technology shock can then be investigated in the light of the nominal model. Moreover, the authors acknowledge the need of a more convoluted bank balance sheet portfolio able to explain further features of the banking sector.

The need to establish optimal bank-size-dependent regulation leads [Goel \(2025\)](#) to develop a banking industry general equilibrium model in which there are incumbent banks and entrant banks. Regulation is regarded to the minimum capital requirement. Differently from [Corbae and D’Erasmus \(2021\)](#), in this model all banks act in a competitive banking sector and therefore, there is not a leader bank and followers banks. While entry into the model is endogenous, the model also abstracts from banks’ market power on credit.

The model underscores the rise in welfare and confirms the fall in credit supply in response to a stricter minimum capital requirement for all banks. The result occurs because bank capital in the economy rises because of the increase in the mass of banks. The fall in credit is however lower than an economy without heterogeneous banks. Analysing the optimal size dependent policy, the author finds that stricter regulation on large banks than small banks increases welfare because it reduces the failure rate and concentration of risk of large banks as well as the number of large and less efficient banks. The result is reinforced when bankruptcy costs are introduced into the model as turnover costs fall. The model is suited for the introduction of sticky price to study the aggregate implications of monetary policy.

Research recently carried out has further established the impact of tighter capital requirements, high entry barriers and compliance costs imposed by Dodd-Frank Act regulation and supported the fall in credit supply and increase in exit rate and lending rate ([Liu, 2025](#)). In contrast to [Goel \(2025\)](#), [Liu \(2025\)](#) emphasises the fall in the number of small banks instead of large banks as the model consists of small banks only. In the long run a positive selection effect is detected since surviving banks expand their size through investment driving down bank costs and increasing concentration. The paper distinguishes itself by performing structural estimation through simulated method of moments with specific focus on non-interest costs that consist of seven parameters to be calibrated with ten moment conditions. The ‘ladder climbing’ process of size expansion is supported by the employment of the probabilistic structure over equity in analogy with [Corbae and D’Erasmus \(2020\)](#) who however depict the climbing process over deposits. While considering loan demand this is entirely exogenous, microfounding the borrower loan demand can help the model quantitatively to match moment counterparts and extend the model to a general equilibrium.

In the models presented in this section (with the exception of [Rios-Rull et al. \(2020\)](#), [Goel \(2025\)](#) and [Liu \(2025\)](#) models) the economy features complete markets, therefore the heterogeneity is explored ex-ante and cannot bring an ex-post distribution in any bank’s dimensions.

²⁰For an example of models with heterogeneous banks in an open economy see [Niepmann \(2023\)](#) who studies banks’ heterogeneity in the efficiency of banks across borders.

The reason is that ex-ante heterogeneity means that banks are unequally exposed to the idiosyncratic shock. As it occurs in [Gertler and Kiyotaki \(2010\)](#) and [Gertler and Karadi \(2011\)](#), the models aggregate since despite the presence of heterogeneous banks the individual state variable is linear and does not display any form of nonlinearity. Moreover, they also assume that ex-ante expected returns to intermediation are equalised across banks at the beginning of each period implying that there is no need to keep track of the distribution of bank network.

5.9 Spatial Heterogeneous Banks

A recent strand of the literature has started to investigate the geographical implications of regulations on banks' performance. [Corbae and D'Erasmus \(2022\)](#) build on [Corbae and D'Erasmus \(2021\)](#) and combine features of [Corbae and D'Erasmus \(2020\)](#) to study the diversification of the idiosyncratic deposit withdrawal through the geographic expansion of commercial banks that while reducing financial instability, it brings more concentration in the industry. The authors focus on the effects of the deregulation followed by the Riegle-Neal Act Interstate Bank Branching Efficiency Act which removed obstacles to banks in opening branches in other US states. The authors find a positive relationship between banks' dimension and diversification. In particular, zooming the diversification effect the volatility of banks' deposits decreases as banks grow in size at regional and national levels.

The economic environment consists of banks Cournot competing within the east and west region. There are three types of banks: state, regional and national banks that—similarly to [Corbae and D'Erasmus \(2020\)](#)—over time expand in size through the exogenous deposit ladder in the Markov process which lowers the volatility of low deposit inflows. The process is able to account for the small mass of large banks with small and regional banks extending less loans than their deposits. Moreover, the model engenders the marginal propensity to lend as the marginal change in profits due to a change in the interest rates. However, parameters are not calibrated in line with this quantity which can provide new insights according to banks' size differentials.

By analysing regional sensitivity, the authors keep constant high aggregate shock, the regional shock in the west region and implement a full regional shock in the east region by setting the aggregate shock equal to the high state for the first five periods, then it decreases to the low aggregate state for three periods and then moves according to its stochastic process. The shock has contractionary effects in lending and output. The main insight is that whilst the west region does not experience any regional shock, it receives spillover effects from the east region where the shock is implemented due to losses and failures of banks. The exit of banks comes primarily from national banks that changes their lending and the industry becomes more concentrated.

[Oberfield et al. \(2024\)](#) also explore the spatial expansion of banks following banks' deregulations in the 1980s and 1990s. They however focus on rationalising two types of sorting: span of control sorting and mismatch sorting through a theoretical model of banks. The 1994 Riegle Neal Act Interstate Bank Branching Efficiency Act removed all restrictions on interstate banks'

locations and gave permission to banks of out of state acquisitions. By 1997 banks' deregulation allowed all commercial banks to open branches across all US states, while in 1981 banks could have operated in their home state only. During this time, the largest banks were set in the most affluent cities, whereas small banks were located in rural locations.

The authors use data from the Federal Deposit Insurance Corporation, the Report of Condition and Income and Census population data from 1980 to 2006. They document some important facts. First, during the deregulation period the number of banks declined by 45% whereas the number of branches increased by 60% with a 57% growth in the average number of branches per county. Second, splitting growth at the intensive and extensive margin, banks in the top 1% of the distribution experienced an increase in branches of about 166% whereas branches in the bottom 50% of the distribution declined by 22%. Third, focussing on wholesale funding made by brokered deposits, time deposits and Federal Funds purchases the authors find that large banks in the top 1% of the distribution have a more skewed distribution than the bottom banks. This fact is consistent with the observation that large banks have a more diversified liability structure made by internal and external equity in contrast to small banks. As banks increased the use of wholesale funding, their profitability fell and began expanding to obtain more retail deposits providing the basis for banks' geographical expansion.

The authors develop a heterogeneous banking model with heterogeneous locations. Households have preferences over branches and distance to a bank branch where they are subject to idiosyncratic shock, while banks make decisions over a branch and then decide for each branch the interest rate on loans and deposits. Any difference between loans and deposits is financed with wholesale funding which is more costly because it is uninsured funding. The optimisation is over the distance to the closest bank branch which together with the constraints give rise to a distribution for loans and deposits. Productive banks are more willing to use wholesale funding since they can take advantage of economies of scale and diversify, thus the more the wholesale funding in a bank balance sheet the larger the bank size and the higher the interest rates on loans and deposits. Banks try to locate closer to the headquarter.

The main theoretical contribution is the introduction of span of control sorting and mismatch for banks. The span of control sorting implies that more productive banks (less productive banks) locate in denser, richer and more expensive (cheaper, less dense and less expensive) locations where there is a high demand for deposits. The mismatch sorting, on the other hand, implies that banks place branches in locations where the funding needs of the bank match with deposits and loans needs of customers with a particular high demand for deposits. This is because banks try to minimise the reliance on wholesale funding which needs to be used when there is a large gap between deposits and loans.

The author find that about 55% of banks belonging to the 1% of the distribution of banks locate in the densest cities, about 15% of these banks are in the bottom three quartiles of counties and small banks locate in less dense cities. Controlling for market size effects during the

deregulation period, the author find evidence of span of control sorting, whereby a 1% increase in banks' size is associated with 33% increase in the average density of banks' counties. While large banks locate in dense location out of state and away from the headquarters creating a mismatch sorting, small banks choose less dense locations. Moreover, a 10% rise in the distance from the headquarters reduces the number of banks by 17-20%, indeed each deregulation analysed in the paper brings a decline in the sorting pattern where large banks expanded only in dense locations they were not yet present. The total decline in the interstate and intrastate sorting is estimated to be 62%. Sorting was relevant for banks before and after the deregulation and became weaker in 2006 than 1981.

Locating in dense locations limits the profitability of large banks since dense locations demand more loans than deposits which raise the exposure of banks to wholesale funding. Since large banks prefer locations with a high demand for deposits, this creates a mismatch sorting because these places are less dense locations. Before the deregulation phase banks with abundant wholesale funding were located in dense locations, after the deregulation the relationship between wholesale funding and density of the county weakened. In particular, after the deregulation phase the relaxation of large banks' liquidity constraints allowed them to increase their deposits, reduce the wholesale funding and expand geographically. Once large banks established their presence in dense locations, they expanded in less dense locations to obtain a higher demand for deposits and taking advantage of the deregulation reducing their cost on wholesale funding, thereby reducing the span of control sorting.

Morelli et al. (2025) develop a spatial banking model where banks compete oligopolistically across multiple geographic regions. Banks do not make a choice over location but they are exposed to local deposit supply shocks. The exposure of banks to the shock depends on the proportion of deposit share in each location. After the implementation of the Riegle-Neal Interstate Banking and Branching Efficiency Act (1994), banks expanded geographically and by consolidation. The authors study this expansion through the market power of banks and its interaction with the funding volatility. In particular, by operating in many locations banks are subject to lower funding risk because of diversification.

The model contains a representative household who makes choices over consumption and liquidity services and a large number of banks who choose the interest rate on deposits. Two environments are proposed. One in which banks choose the same interest rate on deposits called uniform pricing, and one in which banks choose the interest rate on deposits on the basis of the county they operate in. The environment determines the type of markup banks have. The average market share across all counties is the determinant of the markup in uniform pricing, whereas the individual bank's market share in a specific county determines the markup in the local pricing environment.

The main findings suggest a negative effect on deposit spreads following the geographical expansion of banks consequently Riegle-Neal Interstate Banking and Branching Efficiency Act

(1994). This effect accounts for 20% reduction in deposit spreads in the smallest regions. In terms of competition, the results point to a 5% reduction in markup in the smallest counties and an otherwise unchanged markup for largest counties. Altogether the reduction in deposit spreads and markup led smallest counties to be subject to lower risk premia with more bank entry into the county drawing on a beneficial exposure for small counties. By doubling the volatility of demand deposits to analyse the spatial distribution of economic activity, the rise in risk premia and deposit spreads in large counties is directly transmitted to small counties increasing their vulnerability to shocks.

While this review is mainly focussed on the quantitative and theoretical aspects of heterogeneous banks, for completeness of discussion of articles in spatial banking, I present recent empirical studies on spatial heterogeneous banks. [Aguirregabiria et al. \(2016\)](#) and [Aguirregabiria et al. \(2025\)](#) develop structural models to study the implications of the Riegle-Neal Interstate Banking and Branching Efficiency Act (1994) on geographic risk diversification. In the former article, geographic risk diversification opens the possibility for bank to diversify their risk by opening local branches in multiple locations since idiosyncratic risks can be spatially correlated. Tasked with 3,1000 US counties, the authors implement a factor model to estimate the deposit risk at the county level.

A bank that operates in one county would have a 1.1 and 3.1 percentage points of deposit risks for the 10th and 90th percentiles of the banks' distribution of geographic risk. In the latter article, the authors study geographic imbalances of deposits and loans through bank networks and market power. By developing a structural multi market model of banks' competition they provide a new measure of geographic imbalance. The imbalance index is used to measure the degree of segregation in the deposits and loans market, taken as the transfer of funds within branches and across counties.

The results suggest that despite their deposit shares, some regions do not have enough credit opportunities. Yet, the total amount of deposits affects the amount of banks' loans. The presence of market power at local level poses negative effect on the geographic flows of credit and on the social surplus especially for poorer counties.

6 Estimating Heterogeneous Banks' Responses to Monetary Shocks

In this Section I provide estimates for commercial banks in the US to monetary shocks under the circumstance in which negative interest rates had been embarked in the US by the Federal Reserve. Research conducted to date finds that an increase in the federal funds rate leads to a decline in the loan supply of illiquid small banks larger than liquid small banks (Kashyap and Stein, 2000; Jiménez et al., 2012). However, this channel has become obscured post-financial crisis due to changes in the banking sector with the fraction of liquidity constrained small banks increasing and the adoption of negative rates by some monetary authority. The following analysis

explores more deeply the rebalancing of banks' assets and the bank lending channel.

To examine the effect of the banks' rebalancing of assets and the bank lending channel in a negative interest rate environment I estimate the loans supply, represented by banks' illiquid assets at bank year level. In particular, I estimate the following econometric specification:

$$\Delta Illiquidassets_{i,t} = \beta_0 + \beta_1 Liquidassets_{i,t} + \beta_2' \mathbf{X}_{i,t} + \nu_{i,t}, \quad (7)$$

where $\Delta Illiquidassets_{i,t}$ is the illiquid assets growth for bank i in year t computed as the log difference in illiquid assets, and $\nu_{i,t}$ is the idiosyncratic error component. The explanatory variables are the measure of banks' liquid assets, $Liquidassets_{i,t}$, the vector of bank level controls $\mathbf{X}_{i,t}$, which I include to isolate the effect of change in interest rate. It includes banks' loans adjustment costs, the loan to deposits ratio, the interest rate on deposits, banks' size as measured by the log of total assets, tier 1 capital to total assets, the liquidity to deposits ratio, and a second measure of banks' size as measured with log of deposits. The coefficient of interest β_1 is identified by comparing the value in credit growth by banks in the same period with different liquidity measures. Estimation results are computed with panel regression and estimators are corrected for standard errors.

The identifying assumption is that conditional on banks' effects and control variables, banks relying less on deposits provides the lending response of high deposits banks at a time post-financial crisis when the negative interest rate is undertaken. I assume that a negative policy rate is in place post-financial crisis. This identification strategy follows [Heider et al. \(2019\)](#) who study the monetary policy of negative rates in the Euro Area. However, while they posit their identifying assumption within the context of a difference-in-differences estimation in which the negative rates policy is absent, I base my identifying assumption on the fact that the Fed did not adopt negative rates. Therefore, the lending response post-financial crisis is a counterfactual response in the presence of negative rates. Essentially, the analysis provides the response of illiquid assets growth following a monetary changes encoded via different measures.

I introduce control variables to account for the endogeneity of monetary policy and minimise the influence of confounding factors. The main threat to the identification are the time-varying effects across banks. I assess the robustness of the specification by omitting and reintroducing deposits over the estimation windows which is the variable that provides the counterfactual on banks' behavior, by accounting of the size diversity of banks through assets and by using different measures for the variable liquid assets.

Table 3 presents the results for the two samples. One from 1998 to 2008 prior to the financial crisis and, from 1998 to 2016 considering the impact of the financial crisis. I vary the estimation window to minimise the influence of banks' characteristics around the financial crisis. The key explanatory variable is banks' liquid assets. The first measure of liquid assets is a refined measure that I use to control for banks' liquidity positions and to reduce the impact of confounding effects on liquidity. It contains cash and balances from depository institutions,

TABLE 3.—Banks' portfolio rebalancing of assets

Time windows	Illiquid assets growth							
	Liquid assets ₁				Liquid assets ₂			
	1998-2008		1998-2016		1998-2008		1998-2016	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Liquid assets _{<i>i</i>}	-0.109*** (0.000)	-0.021*** (0.000)	-0.101*** (0.000)	-0.015*** (0.000)	-0.055*** (0.000)	-0.027*** (0.000)	-0.014*** (0.009)	-0.003 (0.257)
Loans costs	0.00*** (0.012)	0.001*** (0.000)	0.00 (0.291)	-0.00 (0.390)	0.001*** (0.006)	-0.000*** (0.000)	-0.00 (0.295)	-0.00 (0.395)
Interest deposits	-0.927*** (0.001)	-1.089*** (0.000)	-0.395 (0.152)	-0.795*** (0.000)	-0.436 (0.147)	-0.954*** (0.000)	-0.157 (0.578)	-0.764*** (0.000)
Loan to deposits	-0.001 (0.248)	-0.001*** (0.000)	-0.000 (0.606)	-0.001*** (0.000)	-0.001 (0.373)	-0.001*** (0.000)	-0.000 (0.931)	-0.001*** (0.000)
Liquidity to deposits	0.1435* (0.062)	-0.039*** (0.004)	0.113* (0.065)	-0.061*** (0.004)	0.117** (0.055)	-0.045*** (0.002)	0.078** (0.038)	-0.068*** (0.004)
Tier 1 to assets	-2.572*** (0.000)	-2.001*** (0.000)	-2.143*** (0.000)	-1.758*** (0.000)	-2.540*** (0.000)	-2.000*** (0.000)	-2.097*** (0.000)	-1.748*** (0.000)
Assets	0.084*** (0.000)	0.811*** (0.000)	0.078*** (0.009)	0.802*** (0.000)	0.008** (0.039)	0.805*** (0.000)	0.007*** (0.014)	0.799*** (0.000)
Deposits		-0.841*** (0.000)		-0.833*** (0.000)		-0.849*** (0.000)		-0.841*** (0.000)
Constant	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14, 135	14, 135	17, 487	17, 487	14, 135	14, 135	17, 487	17, 487
Adjusted <i>R</i> ²	0.226	0.805	0.188	0.804	0.184	0.804	0.137	0.803

NOTE. The sample is from 1998 to 2008 in columns 1-2 and 5-6 and from 1998 to 2016 in columns 3-4 and 7-8. The dependent variables is the growth of banks' illiquid assets measured as log difference. *Loans costs* are charge off loans minus recovery loans to illiquid assets. *Interest deposits* is the interest rate on deposits. *Loans to deposits*, *Liquidity to deposits* are ratios of deposits and *Tier 1 to assets* is the ratio of tier 1 capital to total assets. *Assets* and *Deposits* are measured in log. Figures in parenthesis show robust standard errors. Significance is ****p* < 0.01, ***p* < 0.05 and **p* < 0.10. Data source is in appendix C.

federal funds sold and securities purchased, securities held to maturity and trading assets. In both samples I include separately one at a time deposits as explanatory variable.

The coefficient on liquid assets is negative and statistically significant and shows a stable negative effect across all regressions in columns (1) to (4). This suggests that banks with more liquid balance sheets reduce illiquid assets—loan supply—relatively more than banks with less liquidity following an increase in liquid assets. The estimation picks up an important role for deposits, that is a source of liquidity in banks' balance sheet, and while it attenuates most of the variation in illiquid assets coming from transitory changes in liquid assets, the direction of change is unchanged. Isolating the role of deposits, the results suggest a 10.9% and 10.1% reduction in illiquid assets growth pre and post-financial crisis whose impact is strongly statistically significant. The magnitude of the effect in liquid assets becomes smaller, from 2.1% to 1.5%, pre and post-financial crisis, respectively, but remains statistically significant, when deposits are included in the estimation. The contraction in deposits causes a fall in banks'

lending. If the portfolio rebalancing of assets, given a change in the interest rate, is at work and is the main channel affecting banks' illiquid assets, then there should be no difference in sign in the coefficient of liquid assets. Therefore, banks reallocate their assets towards more liquidity.

A potential concern due to the presence of the financial crisis in the time period, is the endogeneity of monetary policy. This could bias the coefficient of liquid assets upward or downward if the impact of monetary policy is correlated with banks' liquidity positions. To address this concern, I extend the measure of liquidity, as described at the beginning of this Section, to include potential liquidity borrowing positions such as mortgage backed securities.

Comparing the second measure of banks' liquid assets in the last four columns of Table 3 with the first measure of liquid assets, the coefficient on banks' liquid assets become larger. This suggests that there might be a possible endogeneity problem that biases the results downward if endogeneity is not controlled for. The results remain statistically significant.

To account for the endogeneity of monetary policy I employ two proxies for monetary policy *i)* the liquidity to deposits ratio and *ii)* the interest rate on deposits. I introduce the liquidity to deposit ratio in the spirit of banks' reserves requirement and therefore any rise provides the grounds for a contractionary monetary policy and viceversa.

Both factors have an important role in determining the growth in illiquid assets. The results are in line with the predictions of the bank lending channel of monetary policy. Indeed, a one percent increase in the liquidity to deposits ratio decreases banks' illiquid assets to 4.5% and 6.8% pre and post-financial crisis, respectively. Likewise, a one percentage increase in the interest rate on deposits leads to a 95.4% and 76.4% fall in banks' illiquid assets pre and post-financial crisis, respectively. While the bank lending channel of monetary policy is at work also in periods of excess liquidity post-financial crisis, its effects attenuate and banks' lending behavior increases, instead of decreasing, relative to the pre-financial crisis period. The fall in lending comes primarily from liquidity constrained banks, *i.e.*, banks with high liquidity to deposits ratio.

To examine the strength of the relationship between illiquid assets with loans adjustment costs and loan to deposits ratio, I extend the specification by including these two factors for robustness.²¹ The loans adjustment costs is insignificant post-financial crisis, but it shows a significant negative relationship with loans supply pre-financial crisis. This is consistent with the prediction of the theory, whereby an increase in loans adjustment cost reduces illiquid assets since it decreases banks' profits. Therefore, banks reallocate assets towards more liquidity. By contrast, the significance of the loan to deposits ratio relies on the inclusion of deposits in the estimation. Banks with not only high loan to deposits ratio but also high deposits, reduce more illiquid assets following a one standard deviation increase in the loan to deposit ratio.

To further explores the portfolio rebalancing of assets, another measure of banks' liquidity could include banks' reserve holdings set by the liquidity requirement. This however requires

²¹In Appendix C I describe the construction of the variables loans adjustment costs and loan to deposits ratio.

extending bank level data on required reserves before the financial crisis since this measure was not available in banks' balance sheet. I do this by following the procedure proposed by Afonso et al. (2019). I reestimate the model with this measure of liquidity and I present the results in Appendix B. The results are in line with the benchmark analysis in Table 3, although liquid assets play a less prominent role.

This analysis documents that banks with high liquid balance sheets reduce more illiquid assets than banks with low liquid balance sheets. This is consistent with early findings of Kashyap and Stein (1995, 2000) on the bank lending channel. Additionally, the effect attenuates post-financial crisis but it remains significant in all samples considered. The analysis puts at centre stage a non trivial role for deposits. Importantly, high deposit banks have reduced by more illiquid assets than low deposits banks. These empirical findings shed further light on the cross sectional heterogeneity surrounding the banking sector and highlight bank size heterogeneity as dominant feature to consider in macroeconomic models.

To conclude this evidence point towards a statistically and significant negative impact on banks' illiquid assets given a change in the liquidity to deposits ratio and following the negative interest rate policy, the effect is larger for banks more reliant on deposits. In line with previous empirical findings on Euro Area banks (Heider et al., 2019), I find that negative interest rate policies have restrictive effects on US banks owing to the reduction of the loan supply and the few available loans being risky. I conclude from this evidence that heterogeneity in banks' deposits drives heterogeneity in banks' size.

7 Directions for Future Research

An extensive empirical literature documents the relevance of uncertainty in driving business cycles and aggregate macroeconomic fluctuations. These studies have reached the agreement that uncertainty at both micro and macro levels is negatively related to the economic activity, specifically it increases during periods of declines in economic activity. However, uncertainty brought into the model through real financial frictions has received little attention.²²

In this section I try to tackle two sets of questions that deal with uncertainty recognised as time variation in the second moment of the stochastic process, both questions represent worthwhile research directions.

The first question pertains microeconomic uncertainty from the heterogeneous banking sector on aggregate levels. In particular, following the global financial crisis the interbank market was subject to adverse disruptions due to banks' reluctance to lend liquidity to their counterparts. On the one hand, lenders for wholesale funding required higher credit risk premia; on the

²²On micro and macro empirical evidence see Schwert (1989), Bloom et al. (2007), Bloom et al. (2018), Berger and Vavra (2018), Storesletten et al. (2004), Guvenen et al. (2014), Caldara et al. (2016). On macro quantitative and theoretical models of uncertainty see Bansal and Yaron (2004), Bloom (2009), Andreasen (2010), Fernández-Villaverde et al. (2011), Andreasen (2012), Caldara et al. (2012), Rudebusch and Swanson (2012). For a literature review see Fernández-Villaverde and Guerrón-Quintana (2020).

other hand banks short of liquidity avoided to demand funds in the interbank market in order to not be perceived as illiquid. Both actions resulted in a drying-up of liquidity and, in the best circumstances, a significant reduction of funds traded in the interbank market and a consequent increase in the interbank market rate.

In line with heterogeneous distributional features, [Altavilla et al. \(2026\)](#) measure the effect of interbank market uncertainty for the Euro Area on lending rate and find that banks located in the 10th percentile of the bank capital distribution reduced the lending rate to firms by more than small banks. The effect on credit supply is then a larger reduction from banks located in the 90th percentile of the capital distribution. Previous empirical studies of the interbank market whose uncertainty was generated by the financial crisis also found a larger reduction in lending from banks with high illiquid assets and viceversa, as examined in [Cornett et al. \(2011\)](#), [Buch et al. \(2015\)](#), [Valencia \(2017\)](#) for the US.

The approach in this framework is to quantify the idiosyncratic uncertainty in the interbank market in a heterogeneous bank model with an occasionally binding constraint on liquidity giving particular importance to the stochastic volatility of the idiosyncratic process, namely to the time variation in the variance of the stochastic process.²³ Liquid assets are disentangled between reserves and wholesale funding similarly to [Bianchi and Bigio \(2022\)](#), however in this framework the interbank market arises exogenously and it is now subject to idiosyncratic volatility. First, how does time varying volatility in the interbank market interact with the banks' funding structure? Second, can the interbank funding volatility be related to the state of the economy through precautionary bank behavior when banks are heterogeneous, namely simultaneously generate downturns and a worsening of credit conditions?

This question can help to tackle the instability in the interbank market. It can explain how changes in the volatility of accessing the interbank market may affect the real macroeconomy through the precautionary bank behavior during the severity of the "Covid-19 recession" in which despite the adopted unconventional monetary policy, banks have not transferred the additional liquidity to the economy. Importantly, it can also help to understand how liquidity frictions act as a propagation mechanism for uncertainty shocks when banks are heterogeneous.

The second question is based on aggregate monetary policy uncertainty giving rise to lower first moment on average outcomes motivated by the time variation in the volatility of macroeconomic variables. In this case, I postulate a mechanism by which countercyclical aggregate uncertainty materialises through the lenses of a heterogeneous banking sector and aim to characterise the dynamic responses in unexpected changes in monetary shocks.

The existing literature on policy uncertainty has focussed on aggregate uncertainty shocks with financial frictions embedded as a wait and see effect, or on non-financial corporations, or change in credit spreads. Among many others, examples in this regard from an empirical

²³Examples of models that have studied occasionally binding constraints in a different framework are [Mendoza \(2010\)](#), [Brunnermeier and Sannikov \(2014\)](#), [Bianchi \(2016\)](#), [Bocola \(2016\)](#), [He and Krishnamurthy \(2019\)](#), [Justiniano et al. \(2019\)](#).

and theoretical perspectives include [Fernández-Villaverde et al. \(2011\)](#), [Born and Pfeifer \(2014\)](#), [Christiano et al. \(2014\)](#), [Gilchrist et al. \(2014\)](#), [Fernández-Villaverde et al. \(2015\)](#), [Caldara et al. \(2016\)](#), [Mumtaz and Surico \(2018\)](#), [Arellano et al. \(2012\)](#), [Mumtaz and Theodoridis \(2020\)](#). Differently from this literature the idea is on financial frictions arising from a financial institution, whereas monetary volatility is incorporated as in [Fernández-Villaverde et al. \(2015\)](#) who however focus on volatility shocks arising from fiscal policy.

The direction is to assess a heterogeneous bank model in a general equilibrium with uncertainty in monetary innovations which generates aggregate risk on the side of the bank. Banks in this model can only save for precautionary reasons through liquid assets in conjunction with having adjustment cost in illiquid assets yielding a financial friction to the model. There might be two instances of monetary policy volatility: either it is a pure uncertainty shock in which the uncertainty is perceived by banks, but never materialises or a realised volatility case both of which may have a different effects on the economic activity as documented in the literature on uncertainty in productivity shocks ([Bloom, 2009](#)). [Bordo et al. \(2016\)](#) empirically study macroeconomic policy uncertainty on banks credit growth and find that uncertainty has a negative effect on loans growth for banks of large dimension, less capitalised and with low liquid assets. The effects through the real economy occurs through a standard bank lending channel.

The difference between a model studying idiosyncratic liquidity uncertainty and one with aggregate monetary uncertainty relies not only on the mechanism, but also on the solution procedure to be adopted. Whilst a model with volatility in idiosyncratic uncertainty can be dealt with the uncertainty going into the aggregate distribution over the individual states regardless of the time variation in the standard deviation of stochastic innovation, a model with volatility in aggregate shocks requires a third order approximation of the level equations or higher order to take proper account of the risk-correction of the conditional variance and in alternative to it, a Taylor projection that combines perturbation and projection methods and delivers accurate solutions in case of large shocks as most macroeconomic models need to assume in order to generate realistic fluctuations.²⁴

In [Gigante \(2026a, 2026b\)](#), I take a direction towards these proposals by embedding idiosyncratic liquidity risk and I develop models where banks are facing choices over liquid and illiquid assets to study the importance of the design of improved banks' regulations. While in [Gigante \(2026b\)](#) I develop a model where the interbank market arises endogenously because of the nature of the designed liquidity regulation, I leave for future research the introduction of stochastic volatility into the model to examine the aforementioned research questions.

²⁴On the matter see [Fernández-Villaverde and Levintal \(2018\)](#).

8 Conclusion

In this paper I have examined the current state of heterogeneous banks models that have been focussing either on the credit risk or on the liquidity risk for a bank on two separate dimensions and I have reviewed the evidence for heterogeneous banks and the heterogeneous transmission of monetary policies based on banks' fundamentals. With the aim of documenting the impact on banks of the recent unconventional monetary policy of negative interest rates, I presented a body of empirical evidence that highlights that negative interest rates had heterogeneous effects on banks. I have provided further empirical estimates on the monetary policy effects in the form of negative interest rates on the rebalancing mechanism of banks' assets. The findings reinforce the view that a negative interest rates policy has negative effects on banks since it hurts banks' loans supply growth.

I hope I have identified a set of theoretical models that have direct relevance to ex-post heterogeneous banks. Whilst each of the model discussed has its unique characteristics, they are all related by a common factor that is most often the type of idiosyncratic shock. Furthermore, they all emphasize the relevance of bank size heterogeneity as pressing characteristic to be accounted for in macroeconomic models. I have proposed a thesis in which liquidity regulations can interact with either idiosyncratic uncertainty or aggregate uncertainty through time varying stochastic volatility which can improve the transient dynamics of the model since it can significantly impact the ergodic distribution.

Although theoretical research has not yet investigated the redistributive consequences of conventional and unconventional monetary policies on heterogeneous banks identified in this paper, I believe that when heterogeneous banks are the focus of investigation controlling for a non-degenerate ex-post distribution of banks is necessary to account for the different behavior across banks and for a large amount of variance in macroeconomic aggregates.

Further research needs to explore the micro features of bank size heterogeneity if we are to better understand why monetary and macroprudential policies have the potential to affect the pool of heterogeneous banks differently and the economy as a whole. One line of further research direction is the integration of liquidity risk with credit risk combined with characteristics of the market along the industrial organisation environment, which can provide new lens through which to study size heterogeneous banks in a context of monetary and macroprudential policies through non convex balance sheets and nonlinear liquidity and credit frictions.

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Appendix to
**“Theories of Bank Heterogeneity, Liquidity, Credit Risk and
Monetary Shocks: A Literature Review”**

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A Taxonomy of Bank Heterogeneity Studies

This Appendix provides a graphical illustration of the studies on bank heterogeneity analysed in the main text. It contextualises these models by providing a taxonomy classified according to macro areas that whilst different amongst them, they have a unifying feature that is the source of idiosyncratic uncertainty.

B Robustness Analysis

In this section I present additional evidence that support the findings presented in Section 6.

The aim of the empirical analysis is not to make causal inference on the coefficients but to argue that liquid assets and illiquid assets are inversely correlated given a monetary change. The robustness analysis entails exploring further this relationship with a more restricted measure of bank liquidity that is required reserves. This measure is useful as it establishes the basis of a minimum amount of reserves as fraction of deposits that is set by a reserve requirement.²⁵

The new measure of liquidity is computed in two steps following Afonso et al. (2019). First I compute net transaction accounts as the difference between total transaction accounts and the amounts due from depository institutions and cash in the process of collection. In a second step I subtract to this amount the balance of vault cash.²⁶ I rerun the regression as in the main text and reestimate the model with this measure of bank liquidity. The only difference is that I neglect the control for liquidity to deposits ratio. This is because liquid assets is now a pure measure of banks' transactions that, in addition to deposits, would make the ratio highly correlated with liquid assets thereby making liquid assets with a less significant role. In the main text this issue does not arise since liquid assets is a more convoluted measure that encompasses the holding of banks' securities which helps to smooth the relationship with deposits.

Table B.1 shows the results. The significance of the estimators are in line with the predictions in the main text when using the extended measures of banks' liquid assets. While liquid assets maintains its sign, their effect becomes less powerful. This feature indicates that liquid assets become independent to illiquid assets when liquid assets are based purely on net transactions.

Consistently with the estimation in Table 1, the introduction of deposits as control variable picks up most of the variation in illiquid assets. The explanatory power of the regression rises to 80% pre-financial crisis, and to 79.7% post-financial crisis. A change in one percent standard deviation in deposits leads to 83.5% and 82.4% reduction in banks' illiquid assets pre and post-

²⁵The Fed disciplines the reserve requirement that commercial banks need to ensure. The reserve requirement varies across banks according to their size where size is measured with the amount of deposits. Large banks are subject to higher reserve requirements than small banks. On March 15, 2020 the Fed announced the reduction of the reserve requirement to 0% for all banks, as a result of the pandemic.

²⁶Specifically, in the first step the data variables I use are total transaction RCON2215 subtracting the variables RCFD0083, RCFD0085 and RCON0020. In the second step, the variable vault cash RCON0080 is subtracted from the amount computed in the first step.

Bank heterogeneity and its outcomes to individual banks' components

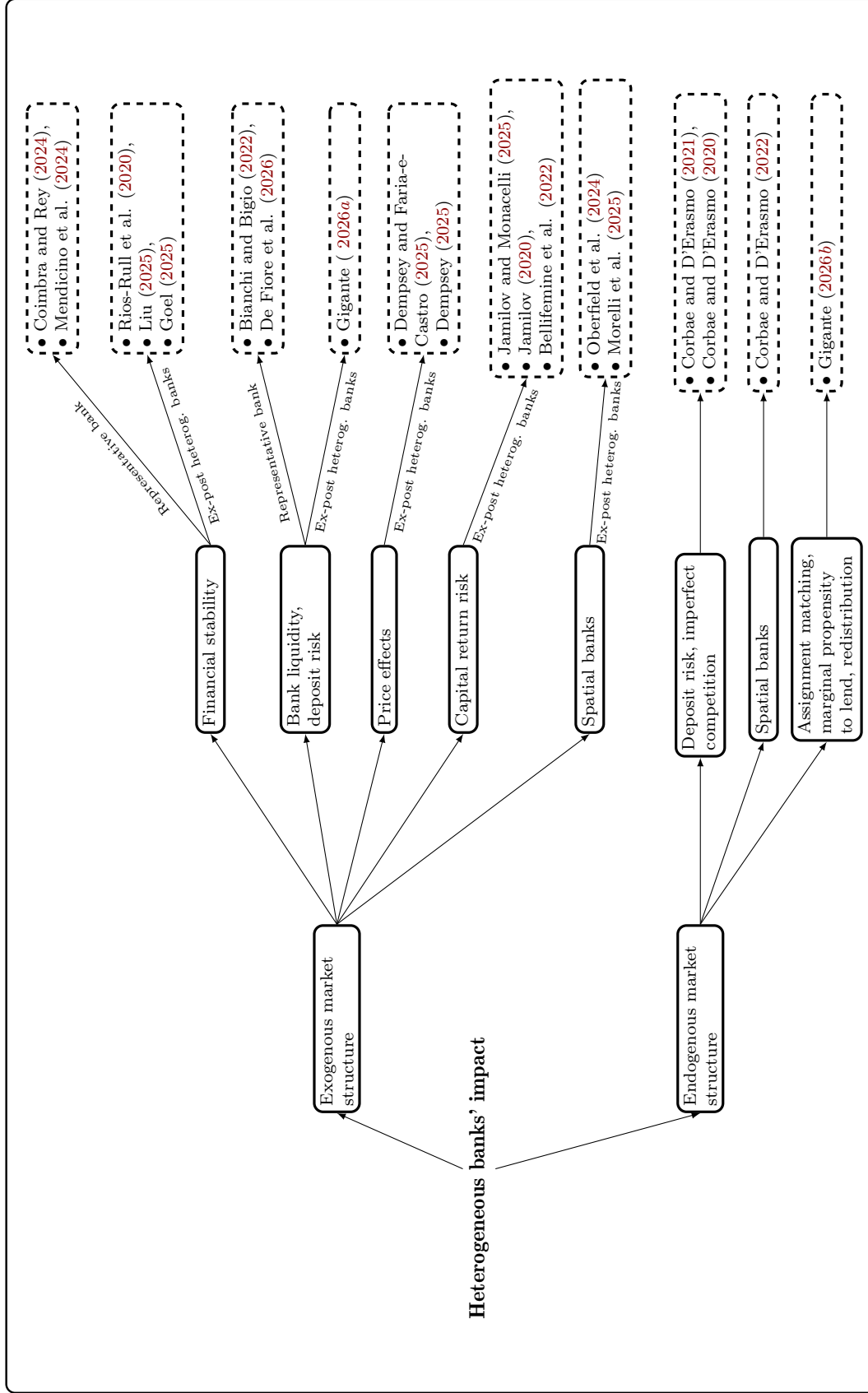


TABLE A.1.—Classification chart of heterogeneous banks' studies in Section 5 of main text.

TABLE B.1.—Robustness analysis required reserves

Time windows	Illiquid assets growth			
	Liquid assets ₃			
	1998-2008		1998-2016	
	(1)	(2)	(3)	(4)
Liquid assets _{<i>i</i>}	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
Loans costs	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.314)	0.000 (0.443)
Interest deposits	-0.663** (0.029)	-1.005*** (0.000)	-0.202 (0.476)	-0.734*** (0.000)
Loan to deposits	0.000 (0.322)	-0.001*** (0.000)	0.000 (0.313)	-0.001*** (0.000)
Tier 1 to assets	-2.379*** (0.000)	-2.039*** (0.000)	-2.013*** (0.000)	-1.819*** (0.000)
Assets	0.006* (0.086)	0.792*** (0.000)	0.007** (0.018)	0.785*** (0.000)
Deposits		-0.835*** (0.000)		-0.824*** (0.000)
Constant	Yes	Yes	Yes	Yes
Observations	14, 135	14, 135	17, 487	17, 487
Adjusted R^2	0.161	0.800	0.130	0.797

NOTE. The sample is from 1998 to 2008 in columns 1-2 and from 1998 to 2016 in columns 3-4. The dependent variables is the growth of banks' illiquid assets measured as log difference. *Loans costs* are charge off loans minus recovery loans to illiquid assets. *Interest deposits* is the interest rate on deposits. *Loans to deposits* is ratio of deposits and *Tier 1 to assets* is the ratio of tier 1 capital to total assets. *Assets* and *Deposits* are measured in log. Figures in parenthesis show robust standard errors. Significance is ***p < 0.01, **p < 0.05 and *p < 0.10. Data source is in appendix C.

financial crisis, respectively. Interest rate on deposits displays a statistically significant impact in magnitude to illiquid assets. Therefore, deposits and the interest rate on deposits play a non negligible role in guiding the change in illiquid assets.

C Data Details

I collect micro banks data for a large panel of banks from 1998 to 2016 using data collected in Gigante (2026b). The data source is the Consolidated Report of Condition and Income submitted by commercial banks to the Federal Reserve provided by Wharton Research Data Services (2023). Data are collected for banks located in the fifty states and the District of Columbia following Corbae and D'Erasmus (2021). I collect variables for *i) Liquid assets*, *ii) Illiquid assets*, *iii) Liabilities* from balance sheet statements; and variables for *Income and cost* from the income statements. I use two measures of bank liquidity. The first measure contains a subset of collected variables in *i)* of the balance sheet, which are RCFD0010 Cash and balances due from depository institutions, RCFD1350 Federal funds sold and securities purchased under agreements to resell in domestic offices of the bank and of its edge and agreement subsidiaries,

RCFD1754 Held to maturity securities, and RCFD3545 Trading assets total.

The second measure contains all variables used for *i) Liquid assets* of the balance sheet variables. Loans adjustment costs are defined as the difference between charge off loans and recovery loans similarly to [Corbae and D’Erasmus \(2021\)](#).

The data used in the robustness analysis in Appendix B contains: RCON2215 Total transaction accounts, RCFD0083 Balances due from US branches and agencies of foreign banks, RCFD0085 Balances due from other depository institutions in the US, RCON0020 Cash items in the process of collection, RCON0080 Currency and coin.